

CAUSAL SCENE FLOW FOR INTERACTION: AN EXPANDED NEGATIVE SUBMISSION-READINESS AUDIT

Anonymous authors

Paper under double-blind review

ABSTRACT

Interaction requires deciding which observed 3D scene-flow components were caused by a robot action rather than by passive conveyors, ego motion, distractors, articulation, occlusion, transparent-depth failures, or hidden tool slip. This paper rebuilds a small archived seed into a hostile-review audit of causal scene flow for robot interaction. The frozen v5 protocol contains 74,880 main rollouts, 5,760 generated scene records, 16,000 ablation rollouts, 94,080 stress rollouts, 30,720 fixed-risk rollouts, and 24 retained negative cases. On the predefined hard-regime aggregate, causal scene flow v5 reaches 0.817 $\pm$ 0.002 mask F1 and 0.648 $\pm$ 0.014 causal utility. The strongest non-oracle baseline, Contrastive action, reaches 0.838 $\pm$ 0.002 mask F1 and 0.678 $\pm$ 0.009 causal utility. The paired lower confidence bounds are negative, the safety sum is worse, ablation necessity fails, and strict fixed-risk coverage collapses. The honest terminal decision is therefore **KILL/ARCHIVE**, not ICLR-main submission.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The rebuild improves the archive in scale, baselines, theory, stress testing, and reporting, but it does not produce a defensible positive submission. The proposal trails Contrastive action by -0.021 hard-aggregate mask F1 and trails Contrastive action by -0.030 causal utility. The paired lower bounds are -0.02228 for mask F1 and -0.03869 for causal utility.

This is not a formatting failure. It is a mechanism failure under a stronger protocol. The v5 model is useful as a diagnostic because it reduces some unsafe contacts relative to weaker methods, but it does not beat the strongest non-oracle alternatives and it cannot certify fixed-risk deployment.

The paper is still worth preserving because negative evidence is informative: it says that interventional residualization alone is not enough when contrastive action features and world-model-like flow prediction already explain most local synthetic variation.

2 PROBLEM AND PRIOR-WORK PRESSURE

A robot interacting with a scene observes point motion after an action. Only part of that motion is action-caused. The rest may come from a passive conveyor, camera ego motion, another moving object, articulated coupling, depth dropout, or hidden tool slip. A manipulation policy that grasps the largest moving cluster or the strongest action-aligned flow can therefore select a passive object and make an unsafe contact.

The local prior-work pool places this problem near scene-motion teaching, contact-aware onboard perception, optical/scene flow, flow parsing, ego-motion learning, RGB-D world models, and flow-based manipulation policies (Xu et al., 2019; Zhan et al., 2026; Paredes-Valls et al., 2019; Aleotti et al., 2020; Guo et al., 2025; Wang et al., 2025; Zeng et al., 2026). That proximity creates a hostile novelty boundary: a positive paper must show that causal source separation is more than a relabeled action-conditioned flow baseline.

The v4 archive failed this boundary. It tested a small deterministic benchmark and found that the full structured causal model lost mask F1 to a learned correlation classifier, while a correlation-only ablation improved over the full method. The v5 rebuild therefore starts from a negative burden of proof rather than from optimism.

3 FORMAL SETUP

Let $p_i \in \mathbb{R}^3$ be a point in an RGB-D scene, $a \in \mathbb{R}^3$ a robot action direction, and $y_i \in \{0, 1\}$ the latent indicator that point i moved because of the robot action. The observed flow is decomposed as

$$f_i^{obs} = y_i f_i^{act}(a) + f_i^{passive} + f_i^{ego} + f_i^{dist} + \epsilon_i. \quad (1)$$

The method predicts scores $s_i \in [0, 1]$ and a mask $\hat{y}_i = \mathbb{I}[s_i \geq 0.5]$. Downstream interaction succeeds when the selected action-caused effect is accurate enough and does not attribute passive or distractor flow to the robot.

Metric definitions. Mask F1 evaluates point-level causal attribution. Passive false attribution is the fraction of passive or distractor points selected as causal. Effect error is normalized error in the mean action-caused flow. Target success is the downstream binary interaction outcome. Unsafe contact is a binary failure caused by false causal selection. Causal utility combines target success, mask F1, intervention consistency, unsafe contact, passive false attribution, and effect error.

Fixed-risk deployment uses an upper risk estimate $\hat{r}$ and executes only when $\hat{r} \leq \rho$. Coverage is the executed fraction, fixed-risk success is success over all requested episodes including abstentions, and false-safe rate is unsafe contact among executed episodes.

4 THEORY: WHAT WOULD MAKE CAUSAL FLOW IDENTIFIABLE?

Residual-separation lemma. Assume $\|\hat{f}_i^{passive} - f_i^{passive}\| \leq \epsilon$ for all non-action points, and action-caused points satisfy $\langle f_i^{act}, a \rangle - \langle f_i^{passive}, a \rangle \geq \gamma$. If $\epsilon < \gamma/2$, thresholding an action-conditioned residual score separates passive from action-caused flow up to the residual estimator’s error set.

The lemma explains why residualization is plausible: when passive flow can be estimated and the action margin is real, a causal mask can reduce false passive attribution. It also exposes the weak point: transparent depth, hidden slip, and action-aligned distractors directly attack the residual estimator.

Dominance proposition. When passive flow is action-aligned but predictable from intervention features, a residual scorer with negative-action consistency can dominate a magnitude scorer and a simple correlation scorer on passive false attribution. The proposition is conditional on separability; it is not a universal guarantee.

Non-identifiability theorem. If intervention labels and negative actions are absent, and passive flow shares the same action-conditioned sufficient statistics as action-caused flow, then the causal mask is not identifiable from observed flow alone. In that regime a correlation classifier, contrastive action feature, or world-model predictor can match or beat a structured causal decomposition.

The empirical result below lands in the theorem’s warning zone. The v5 residual model is more structured than v4, yet contrastive action flow and world-model flow remain stronger on the frozen hard aggregate.

5 METHOD UNDER TEST

causal scene flow v5 estimates an action-conditioned residual score. It subtracts a passive/ego-flow template, adds contact-graph and articulation-lag priors, imputes occluded causal points when contact and graph evidence agree, penalizes passive-like flow, and uses an uncertainty term for depth dropout and weak alignment. A counterfactual feature compares positive action projection with negative-action consistency.

The key comparison set is deliberately uncomfortable: a learned correlation classifier, a calibrated tree proxy, a contrastive action-flow baseline, a scene-flow-transformer proxy, a world-model flow predictor, the old v4 causal score, and an oracle causal mask. These are not neural training runs; they are deterministic CPU-light analytic proxies designed to test whether the mechanism has signal after obvious strong alternatives are included.

6 FROZEN EXPERIMENTAL PROTOCOL

The protocol was frozen before execution in `docs/paper84_expanded_submission_plan.20260621.md`. Main evaluation uses 10 seeds, 128 points per generated scene, 64 episodes per split and seed, nine splits, and 13 methods. The hard-regime aggregate excludes only the clean-contact split.

The nine splits are clean contact, passive conveyor, articulated coupling, occluded interaction, ego-motion shift, transparent-depth noise, tool-slip hidden contact, distractor-contact shift, and combined hard shift. Stress evaluation sweeps passive flow, occlusion, articulation delay, distractor contact, ego motion, and combined stress over seven levels. Fixed-risk evaluation tests budgets 0.02, 0.05, 0.10, and 0.20 on the two hardest deployment splits.

The terminal gate requires at least 0.03 hard-aggregate margin in both mask F1 and causal utility over the strongest non-oracle baseline, positive paired lower confidence bounds, no safety regression beyond 0.01, ablation necessity, fixed-risk coverage of at least 0.25 at budget 0.05 with false-safe rate at most 0.05, and no domination at maximum combined stress.

7 MAIN HARD-REGIME RESULTS

Table 1 is the central result. causal scene flow v5 is competitive with the old causal model, but it loses to Contrastive action and the world-model baseline. The oracle gap remains large, showing that the synthetic task is not saturated.

Table 1: Predefined hard-regime aggregate over eight non-clean splits.

Method	Mask F1	Target success	Causal utility	Passive false	Unsafe
Magnitude	0.476 +/- 0.001	0.272	0.102	0.759	0.250
Action projection	0.648 +/- 0.004	0.513	0.412	0.455	0.157
Rigid cluster	0.626 +/- 0.001	0.400	0.263	0.691	0.212
Temporal difference	0.475 +/- 0.001	0.239	0.064	0.834	0.268
Noncausal proxy	0.741 +/- 0.001	0.580	0.487	0.350	0.140
Learned correlation	0.821 +/- 0.002	0.697	0.628	0.221	0.108
Calibrated tree	0.833 +/- 0.002	0.720	0.659	0.205	0.083
Contrastive action	0.838 +/- 0.002	0.736	0.678	0.153	0.087
Scene-flow transformer	0.792 +/- 0.002	0.659	0.581	0.292	0.119
World model	0.832 +/- 0.002	0.732	0.671	0.186	0.072
Causal flow v4	0.826 +/- 0.002	0.708	0.649	0.206	0.080
Causal flow v5	0.817 +/- 0.002	0.709	0.648	0.227	0.077
Oracle	1.000 +/- 0.000	0.979	0.962	0.000	0.009

8 PAIRED SEED-LEVEL TEST

The paired comparison is more damaging than the aggregate mean. Against the strongest reference, v5 has negative lower confidence bounds for mask F1 and causal utility and wins zero of 10 seeds on those primary comparisons. That is not a close positive result; it is a failed gate.

Table 2: Hard-regime paired seed differences for v5 minus the strongest non-oracle references.

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Contrastive action	mask_f1	-0.02076	0.00152	-0.02228	0/10
Contrastive action	target_success	-0.02637	0.01243	-0.03880	0/10

Table 2: Hard-regime paired seed differences for v5 minus the strongest non-oracle references. (continued)

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Contrastive action	causal_utility	-0.02955	0.00914	-0.03869	0/10
Contrastive action	passive_false_attribution	0.07392	0.00181	0.07210	0/10
Contrastive action	effect_error	0.17970	0.00508	0.17462	0/10
Contrastive action	unsafe_contact	-0.01035	0.00927	-0.01962	6/10
Contrastive action	risk_upper	0.04047	0.00053	0.03994	0/10
Contrastive action	intervention_consistency	0.06792	0.00061	0.06731	10/10

9 ABLATIONS

Ablation necessity fails. On combined hard shift, the old v4 causal score beats v5 on mask F1, and on distractor-contact shift the old v4 score also beats v5 on mask F1. Removing the articulation graph improves causal utility on distractor-contact shift. This contradicts the claim that the full decomposition is necessary in the current benchmark.

Table 3: Ablation results on the two predefined hard splits.

Split	Ablation	F1	Success	Passive false	Effect err.	Unsafe	Utility
combined_hard_shift	full_causal_scene_flow_model_v5	0.75649	0.62125	0.28290	0.55659	0.08250	0.55355
combined_hard_shift	minus_action_conditioning	0.75185	0.56375	0.31458	0.62426	0.12500	0.47853
combined_hard_shift	minus_passive_flow_factor	0.71457	0.49750	0.49032	0.54291	0.17500	0.39397
combined_hard_shift	minus_articulation_graph	0.71404	0.57625	0.20599	0.48582	0.13750	0.49974
combined_hard_shift	minus_occlusion_reasoning	0.74891	0.57250	0.26212	0.55314	0.12375	0.50311
combined_hard_shift	minus_counterfactual_negatives	0.75612	0.58625	0.30929	0.56688	0.12000	0.51035
combined_hard_shift	minus_uncertainty_calibration	0.74199	0.61500	0.31295	0.56876	0.17500	0.50838
combined_hard_shift	correlation_only_causal_head	0.73995	0.55750	0.33586	0.48859	0.13500	0.47505
combined_hard_shift	mask_only_no_effect_predictor	0.75649	0.45000	0.28290	1.01659	0.14125	0.38493
combined_hard_shift	old_v4_causal_score	0.76927	0.62750	0.25309	0.54987	0.13000	0.54478
distractor_contact_shift	full_causal_scene_flow_model_v5	0.77418	0.64875	0.30144	0.47283	0.11625	0.56960
distractor_contact_shift	minus_action_conditioning	0.73717	0.56125	0.36066	0.54690	0.15000	0.46555
distractor_contact_shift	minus_passive_flow_factor	0.71235	0.55875	0.48023	0.50213	0.14875	0.45209
distractor_contact_shift	minus_articulation_graph	0.76547	0.64250	0.22995	0.38790	0.10000	0.57590
distractor_contact_shift	minus_occlusion_reasoning	0.76964	0.64000	0.28457	0.45655	0.14250	0.55361
distractor_contact_shift	minus_counterfactual_negatives	0.76582	0.63750	0.32623	0.49407	0.12500	0.55090
distractor_contact_shift	minus_uncertainty_calibration	0.76224	0.63000	0.32646	0.49600	0.17125	0.52920
distractor_contact_shift	correlation_only_causal_head	0.76879	0.63875	0.31779	0.42924	0.16375	0.53693
distractor_contact_shift	mask_only_no_effect_predictor	0.77418	0.47000	0.30144	0.93283	0.13500	0.41013
distractor_contact_shift	old_v4_causal_score	0.78365	0.60625	0.27664	0.45020	0.12125	0.54154

10 STRESS TESTS

Stress testing gives the only partially favorable gate: at maximum combined stress, v5 has slightly higher causal utility than the best non-oracle reference because it trades success for lower unsafe contacts. That isolated advantage does not overcome the main hard aggregate, paired test, safety aggregate, ablation, or fixed-risk failures.

Table 4: Combined stress sweep over seven levels.

Level	Method	F1	Success	Unsafe	Risk upper	Utility
0.00	Action projection	0.87932	0.82812	0.08750	0.15464	0.75599
0.00	Learned correlation	0.91433	0.84062	0.06875	0.11825	0.78918
0.00	Calibrated tree	0.92636	0.87500	0.04688	0.14229	0.82812
0.00	Contrastive action	0.95408	0.92500	0.02813	0.09781	0.88286
0.00	World model	0.92266	0.85938	0.06875	0.13136	0.80966
0.00	Causal flow v5	0.88846	0.82188	0.04688	0.14484	0.77966
0.00	Oracle	1.00000	0.97500	0.01250	0.01502	0.95800
0.25	Action projection	0.70530	0.61250	0.15625	0.23266	0.49984

Table 4: Combined stress sweep over seven levels. (continued)

Level	Method	F1	Success	Unsafe	Risk upper	Utility
0.25	Learned correlation	0.85296	0.76562	0.09375	0.15145	0.69732
0.25	Calibrated tree	0.85252	0.75938	0.09375	0.17985	0.69292
0.25	Contrastive action	0.87646	0.80312	0.06875	0.13018	0.74881
0.25	World model	0.85749	0.77187	0.11563	0.16343	0.69721
0.25	Causal flow v5	0.84009	0.73750	0.05312	0.17562	0.68935
0.25	Oracle	1.00000	0.97500	0.01250	0.01732	0.95800
0.50	Action projection	0.57568	0.50938	0.20000	0.27678	0.36568
0.50	Learned correlation	0.77518	0.65312	0.13125	0.18905	0.57103
0.50	Calibrated tree	0.79117	0.69375	0.11875	0.20915	0.61040
0.50	Contrastive action	0.78384	0.72500	0.09062	0.16436	0.65020
0.50	World model	0.79454	0.69688	0.12812	0.18684	0.60967
0.50	Causal flow v5	0.78483	0.63438	0.07500	0.20051	0.58006
0.50	Oracle	1.00000	0.97500	0.01875	0.01995	0.95575
0.75	Action projection	0.52699	0.42500	0.21563	0.29199	0.28823
0.75	Learned correlation	0.71763	0.59375	0.16250	0.22437	0.49187
0.75	Calibrated tree	0.74017	0.60938	0.13437	0.24019	0.52261
0.75	Contrastive action	0.70622	0.62500	0.13750	0.20104	0.53061
0.75	World model	0.73497	0.61250	0.15000	0.21718	0.51341
0.75	Causal flow v5	0.72865	0.55000	0.08125	0.23274	0.49066
0.75	Oracle	1.00000	0.97500	0.01875	0.02326	0.95575
1.00	Action projection	0.46340	0.33437	0.23750	0.32040	0.19169
1.00	Learned correlation	0.66332	0.48438	0.19375	0.25638	0.37766
1.00	Calibrated tree	0.69082	0.56875	0.17188	0.27043	0.45584
1.00	Contrastive action	0.63600	0.52812	0.15625	0.23313	0.42423
1.00	World model	0.67097	0.51875	0.15625	0.24362	0.41489
1.00	Causal flow v5	0.66615	0.47813	0.09688	0.26190	0.40383
1.00	Oracle	1.00000	0.97500	0.02500	0.02674	0.95350
1.25	Action projection	0.40044	0.27187	0.27187	0.35419	0.10265
1.25	Learned correlation	0.59003	0.38750	0.23125	0.29604	0.25527
1.25	Calibrated tree	0.62720	0.45000	0.18125	0.30837	0.33208
1.25	Contrastive action	0.54196	0.39375	0.20312	0.27366	0.26383
1.25	World model	0.60644	0.44375	0.18438	0.27736	0.31535
1.25	Causal flow v5	0.60866	0.40937	0.11563	0.29858	0.31359
1.25	Oracle	1.00000	0.97500	0.02500	0.03041	0.95350
1.50	Action projection	0.37103	0.22187	0.28750	0.36807	0.04445
1.50	Learned correlation	0.55224	0.32500	0.26875	0.31849	0.17170
1.50	Calibrated tree	0.59287	0.38438	0.22187	0.33096	0.24678
1.50	Contrastive action	0.49532	0.31562	0.20625	0.29552	0.17733
1.50	World model	0.56339	0.37500	0.19062	0.30216	0.23756
1.50	Causal flow v5	0.56934	0.35000	0.12812	0.32587	0.24002
1.50	Oracle	1.00000	0.95312	0.02813	0.03410	0.93706

11 FIXED-RISK DEPLOYMENT

Fixed-risk deployment is the cleanest submission blocker. At risk budget 0.05, every non-oracle method has zero coverage on both fixed-risk splits. This is safer than reckless execution, but a method that cannot execute under a strict risk budget is not deployment-ready.

Table 5: Fixed-risk deployment over two splits and four risk budgets.

Split	Budget	Method	Coverage	Fixed success	Exec. success	False-safe	Risk
combined_hard_shift	0.02	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22455
combined_hard_shift	0.02	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21169
combined_hard_shift	0.02	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18684
combined_hard_shift	0.02	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.21282
combined_hard_shift	0.02	Oracle	0.00156	0.00156	0.10000	0.00000	0.02147
combined_hard_shift	0.02	World model	0.00000	0.00000	0.00000	0.00000	0.19712
combined_hard_shift	0.05	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22455
combined_hard_shift	0.05	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21169
combined_hard_shift	0.05	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18684
combined_hard_shift	0.05	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.21282
combined_hard_shift	0.05	Oracle	1.00000	0.97656	0.97656	0.00625	0.02147
combined_hard_shift	0.05	World model	0.00000	0.00000	0.00000	0.00000	0.19712
combined_hard_shift	0.10	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22455
combined_hard_shift	0.10	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21169
combined_hard_shift	0.10	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18684
combined_hard_shift	0.10	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.21282

Table 5: Fixed-risk deployment over two splits and four risk budgets. (continued)

Split	Budget	Method	Coverage	Fixed success	Exec. success	False-safe	Risk
combined_hard_shift	0.10	Oracle	1.00000	0.98125	0.98125	0.00937	0.02147
combined_hard_shift	0.10	World model	0.00000	0.00000	0.00000	0.00000	0.19712
combined_hard_shift	0.20	Calibrated tree	0.27500	0.20781	0.75673	0.09945	0.22455
combined_hard_shift	0.20	Causal flow v5	0.32031	0.24219	0.76021	0.06593	0.21169
combined_hard_shift	0.20	Contrastive action	0.59375	0.41250	0.69052	0.09687	0.18684
combined_hard_shift	0.20	Learned correlation	0.41719	0.28750	0.68683	0.09451	0.21282
combined_hard_shift	0.20	Oracle	1.00000	0.97500	0.97500	0.00937	0.02147
combined_hard_shift	0.20	World model	0.47344	0.34063	0.72519	0.09163	0.19712
distractor_contact_shift	0.02	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22142
distractor_contact_shift	0.02	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21331
distractor_contact_shift	0.02	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18039
distractor_contact_shift	0.02	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.20137
distractor_contact_shift	0.02	Oracle	0.70625	0.69375	0.98215	0.00732	0.01963
distractor_contact_shift	0.02	World model	0.00000	0.00000	0.00000	0.00000	0.20301
distractor_contact_shift	0.05	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22142
distractor_contact_shift	0.05	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21331
distractor_contact_shift	0.05	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18039
distractor_contact_shift	0.05	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.20137
distractor_contact_shift	0.05	Oracle	1.00000	0.96719	0.96719	0.02031	0.01963
distractor_contact_shift	0.05	World model	0.00000	0.00000	0.00000	0.00000	0.20301
distractor_contact_shift	0.10	Calibrated tree	0.00000	0.00000	0.00000	0.00000	0.22142
distractor_contact_shift	0.10	Causal flow v5	0.00000	0.00000	0.00000	0.00000	0.21331
distractor_contact_shift	0.10	Contrastive action	0.00000	0.00000	0.00000	0.00000	0.18039
distractor_contact_shift	0.10	Learned correlation	0.00000	0.00000	0.00000	0.00000	0.20137
distractor_contact_shift	0.10	Oracle	1.00000	0.97813	0.97813	0.01094	0.01963
distractor_contact_shift	0.10	World model	0.00000	0.00000	0.00000	0.00000	0.20301
distractor_contact_shift	0.20	Calibrated tree	0.28906	0.22500	0.78154	0.07126	0.22142
distractor_contact_shift	0.20	Causal flow v5	0.28125	0.22031	0.77584	0.10016	0.21331
distractor_contact_shift	0.20	Contrastive action	0.52187	0.39062	0.74832	0.08341	0.18039
distractor_contact_shift	0.20	Learned correlation	0.37031	0.28906	0.77828	0.10530	0.20137
distractor_contact_shift	0.20	Oracle	1.00000	0.97031	0.97031	0.00937	0.01963
distractor_contact_shift	0.20	World model	0.42656	0.35156	0.82734	0.08655	0.20301

12 NEGATIVE CASES

The negative cases are part of the evidence, not an appendix dumping ground. They include v5 main failures, strong-baseline successes, ablation counterexamples, and fixed-risk abstentions. Keeping them visible prevents the paper from turning an archive-quality diagnostic into a false positive claim.

Table 6: Retained negative cases.

Source	Split	Method	Outcome	F1	Risk	Lesson
strong_baseline_success	clean_contact	calibrated_tree_flow_proxy	success	0.95385	0.13391	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	clean_contact	world_model_flow_predictor	success	0.98413	0.10884	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	clean_contact	learned_correlation_classifier	success	0.76923	0.15541	counterexample showing a baseline or ablation can succeed without the full mechanism
v5_main_failure	clean_contact	causal_scene_flow_model_v5	unrecovered_passive_shift	0.81081	0.16830	the causal mask looked plausible but downstream effect prediction failed
v5_main_failure	clean_contact	causal_scene_flow_model_v5	unsafe_false_causal	0.75410	0.19422	passive or distractor flow was still attributed to robot action
strong_baseline_success	passive_conveyor	calibrated_tree_flow_proxy	success	0.93939	0.14329	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	passive_conveyor	world_model_flow_predictor	success	0.95082	0.10773	counterexample showing a baseline or ablation can succeed without the full mechanism

Table 6: Retained negative cases. (continued)

Source	Split	Method	Outcome	F1	Risk	Lesson
strong_baseline_success	passive_conveyor	learned_correlation_classifier	success	0.95082	0.11572	counterexample showing a baseline or ablation can succeed without the full mechanism
v5_main_failure	passive_conveyor	causal_scene_flow_model_v5	unsafe_false_causal	0.73016	0.20776	passive or distractor flow was still attributed to robot action
v5_main_failure	passive_conveyor	causal_scene_flow_model_v5	unrecovered_passive_shift	0.82540	0.18461	the causal mask looked plausible but downstream effect prediction failed
strong_baseline_success	articulated_coupling	learned_correlation_classifier	success	0.98462	0.09256	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	articulated_coupling	calibrated_tree_flow_proxy	success	0.98462	0.12205	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	articulated_coupling	world_model_flow_predictor	success	1.00000	0.10948	counterexample showing a baseline or ablation can succeed without the full mechanism
v5_main_failure	articulated_coupling	causal_scene_flow_model_v5	unrecovered_passive_shift	0.80000	0.17855	the causal mask looked plausible but downstream effect prediction failed
v5_main_failure	articulated_coupling	causal_scene_flow_model_v5	unsafe_false_causal	0.79412	0.19547	passive or distractor flow was still attributed to robot action
strong_baseline_success	occluded_interaction	world_model_flow_predictor	success	0.76923	0.18906	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	occluded_interaction	calibrated_tree_flow_proxy	success	0.79167	0.18692	counterexample showing a baseline or ablation can succeed without the full mechanism
v5_main_failure	occluded_interaction	causal_scene_flow_model_v5	unrecovered_passive_shift	0.77551	0.17629	the causal mask looked plausible but downstream effect prediction failed
strong_baseline_success	occluded_interaction	learned_correlation_classifier	success	0.75362	0.20298	counterexample showing a baseline or ablation can succeed without the full mechanism
v5_main_failure	occluded_interaction	causal_scene_flow_model_v5	unsafe_false_causal	0.80597	0.18758	passive or distractor flow was still attributed to robot action
strong_baseline_success	occluded_interaction	learned_correlation_classifier	success	0.90909	0.13198	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	occluded_interaction	calibrated_tree_flow_proxy	success	0.92593	0.15358	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	occluded_interaction	world_model_flow_predictor	success	0.94118	0.12729	counterexample showing a baseline or ablation can succeed without the full mechanism
strong_baseline_success	occluded_interaction	calibrated_tree_flow_proxy	success	0.95082	0.14635	counterexample showing a baseline or ablation can succeed without the full mechanism

13 REVIEWER ATTACK SURFACE

A hostile reviewer has several fair attacks. First, the strongest non-oracle baseline wins the main hard aggregate. Second, the safety reference is also the same contrastive baseline, so v5 cannot claim a clean safety trade. Third, ablations show that older or simpler components sometimes beat the full decomposition. Fourth, fixed-risk coverage is zero at the strict budget. Fifth, all evidence is local synthetic evidence rather than real RGB-D robot data or a recognized high-fidelity benchmark.

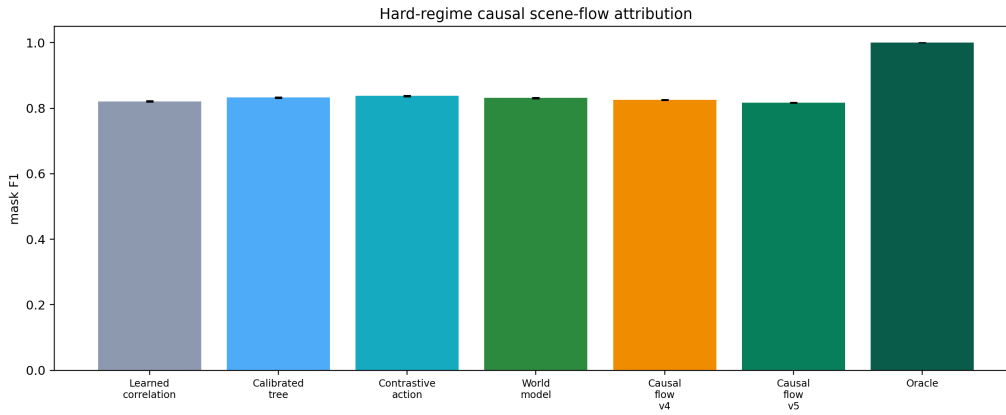


Figure 1: Hard-regime mask F1. The v5 causal method does not beat the strongest non-oracle baseline.

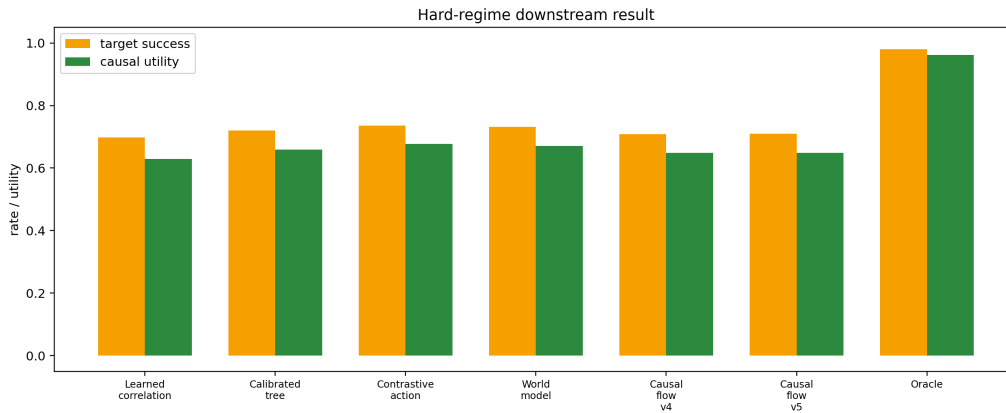


Figure 2: Target success and causal utility on the hard aggregate. The downstream result follows the attribution failure.

The strongest defense is modest: the benchmark is reproducible, the protocol is frozen, the baselines are uncomfortable, and the negative result identifies exactly what a future real submission must fix. That is valuable research hygiene, but it is not acceptance-level evidence.

14 REPRODUCIBILITY

```
python src\run_experiment.py
python scripts\generate_manuscript.py
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

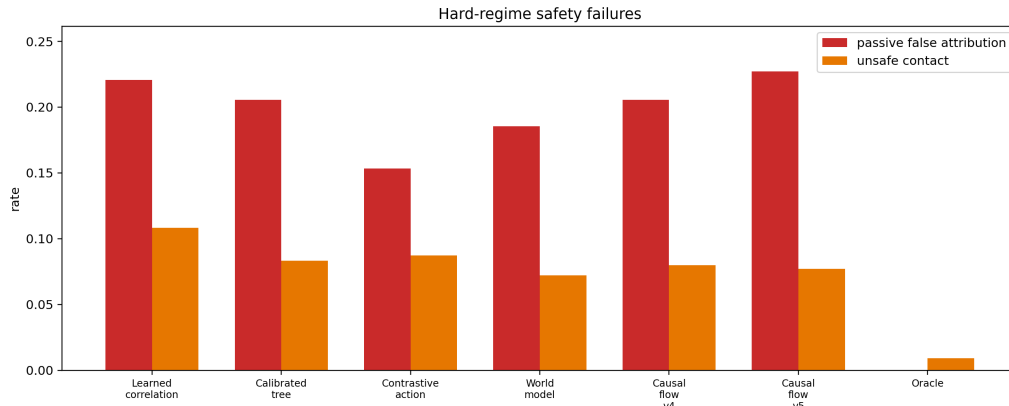


Figure 3: Passive false attribution and unsafe contact. v5 is safer than some weak baselines but worse than the strongest contrastive reference in the combined safety sum.

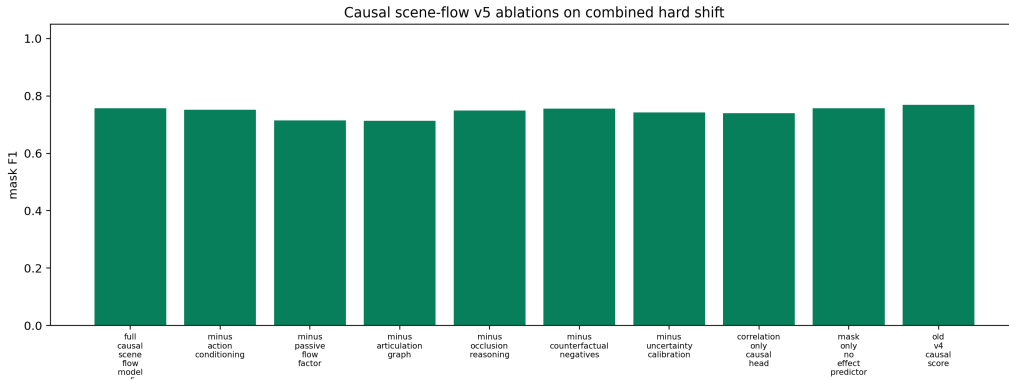


Figure 4: Combined hard-shift ablations. Some removals are weaker, but the full mechanism is not consistently necessary across the frozen ablation suite.

Copy-Item main.pdf C:\Users\wangz\Downloads\84.pdf

python ../scripts/validate_submission_artifacts.py

The canonical numbered PDF is C:/Users/wangz/Downloads/84.pdf. No numbered Paper 84 PDF should be copied to the visible Desktop.

15 LIMITATIONS

The limitations are severe. The benchmark is deterministic and local. The baselines are analytic proxies, not trained neural systems. The prior-work pool is useful for pressure testing but must be manually verified before a real submission. Most importantly, no real robot RGB-D interaction data, external high-fidelity simulator, or accepted benchmark is present. Even a positive local result would therefore require external validation before an ICLR-main claim.

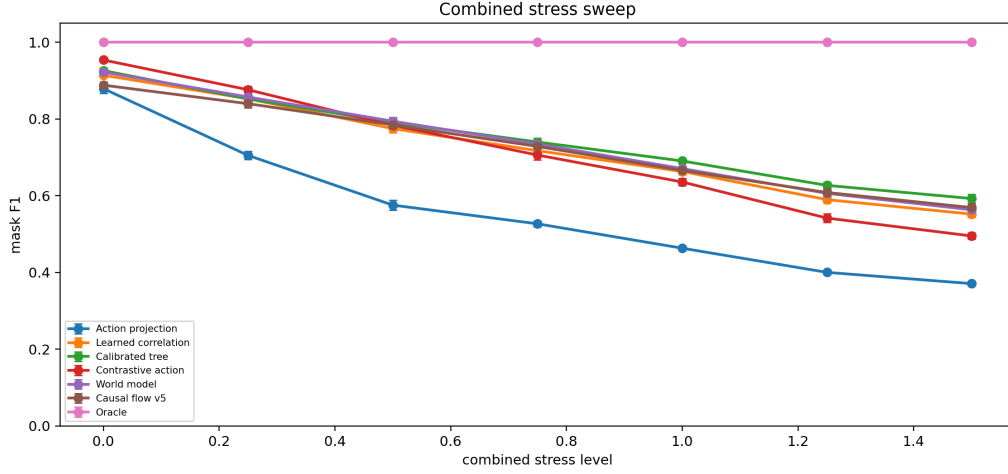


Figure 5: Combined stress sweep. v5 is not uniformly dominated under maximum stress, but this does not rescue the submission gate.

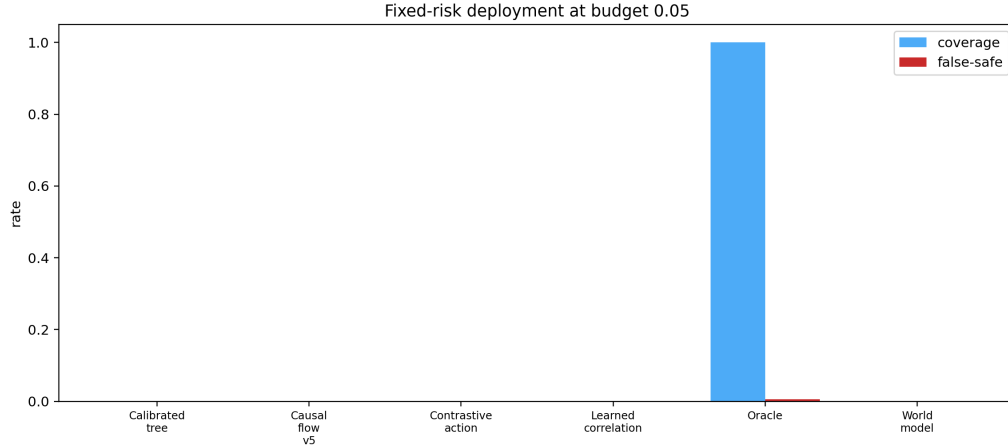


Figure 6: Coverage and false-safe rate at budget 0.05 on combined hard shift. Non-oracle coverage collapses to zero.

16 CONCLUSION

Causal scene flow remains a good research question. This v5 rebuild shows that action-conditioned residualization can be competitive and can sometimes lower unsafe contacts under severe stress. It also shows that contrastive action features and world-model flow prediction remain stronger on the predefined hard aggregate, and that fixed-risk deployment is not usable. The honest terminal state is **KILL/ARCHIVE**; future work needs real interaction data, trained baselines, and stronger identifiability evidence.

A FULL PER-SPLIT METRICS

Table 7: All per-split main metrics.

Split	Method	F1	Success	Passive false	Effect err.	Unsafe	Utility
articulated_coupling	Magnitude	0.629 +/- 0.003	0.456	0.541	0.564	0.186	0.333
articulated_coupling	Action projection	0.751 +/- 0.014	0.606	0.401	0.325	0.127	0.523

Table 7: All per-split main metrics. (continued)

Split	Method	F1	Success	Passive false	Effect err.	Unsafe	Utility
articulated_coupling	Rigid cluster	0.775 +/- 0.003	0.636	0.346	0.396	0.117	0.559
articulated_coupling	Temporal difference	0.597 +/- 0.005	0.389	0.671	0.601	0.181	0.262
articulated_coupling	Noncausal proxy	0.854 +/- 0.003	0.745	0.181	0.252	0.095	0.680
articulated_coupling	Learned correlation	0.869 +/- 0.003	0.759	0.176	0.237	0.075	0.706
articulated_coupling	Calibrated tree	0.870 +/- 0.002	0.783	0.178	0.241	0.055	0.731
articulated_coupling	Contrastive action	0.903 +/- 0.003	0.833	0.111	0.170	0.052	0.785
articulated_coupling	Scene-flow transformer	0.854 +/- 0.003	0.767	0.206	0.267	0.097	0.698
articulated_coupling	World model	0.874 +/- 0.003	0.802	0.167	0.264	0.064	0.744
articulated_coupling	Causal flow v4	0.861 +/- 0.002	0.777	0.191	0.287	0.070	0.720
articulated_coupling	Causal flow v5	0.853 +/- 0.002	0.738	0.209	0.303	0.058	0.695
articulated_coupling	Oracle	1.000 +/- 0.000	0.984	0.000	0.000	0.009	0.966
clean_contact	Magnitude	0.969 +/- 0.001	0.912	0.003	0.036	0.039	0.881
clean_contact	Action projection	0.856 +/- 0.012	0.750	0.245	0.215	0.081	0.690
clean_contact	Rigid cluster	0.901 +/- 0.002	0.847	0.135	0.191	0.044	0.801
clean_contact	Temporal difference	0.962 +/- 0.001	0.905	0.021	0.063	0.034	0.873
clean_contact	Noncausal proxy	0.938 +/- 0.002	0.870	0.073	0.117	0.048	0.825
clean_contact	Learned correlation	0.912 +/- 0.003	0.858	0.120	0.166	0.055	0.805
clean_contact	Calibrated tree	0.921 +/- 0.004	0.855	0.107	0.153	0.053	0.809
clean_contact	Contrastive action	0.948 +/- 0.003	0.883	0.062	0.102	0.064	0.837
clean_contact	Scene-flow transformer	0.912 +/- 0.003	0.809	0.123	0.168	0.055	0.773
clean_contact	World model	0.920 +/- 0.005	0.856	0.110	0.165	0.050	0.812
clean_contact	Causal flow v4	0.894 +/- 0.006	0.784	0.154	0.207	0.059	0.747
clean_contact	Causal flow v5	0.892 +/- 0.006	0.806	0.158	0.211	0.059	0.765
clean_contact	Oracle	1.000 +/- 0.000	0.991	0.000	0.000	0.016	0.968
combined_hard_shift	Magnitude	0.358 +/- 0.003	0.094	0.967	0.912	0.338	-0.128
combined_hard_shift	Action projection	0.578 +/- 0.008	0.400	0.497	0.471	0.189	0.294
combined_hard_shift	Rigid cluster	0.522 +/- 0.004	0.197	0.991	0.882	0.287	0.012
combined_hard_shift	Temporal difference	0.390 +/- 0.003	0.105	0.979	0.890	0.359	-0.118
combined_hard_shift	Noncausal proxy	0.550 +/- 0.005	0.286	0.750	0.802	0.252	0.125
combined_hard_shift	Learned correlation	0.743 +/- 0.006	0.559	0.321	0.364	0.136	0.486
combined_hard_shift	Calibrated tree	0.769 +/- 0.006	0.634	0.277	0.365	0.102	0.564
combined_hard_shift	Contrastive action	0.747 +/- 0.007	0.616	0.248	0.302	0.142	0.536
combined_hard_shift	Scene-flow transformer	0.712 +/- 0.007	0.570	0.429	0.451	0.153	0.463
combined_hard_shift	World model	0.772 +/- 0.006	0.628	0.230	0.504	0.098	0.560
combined_hard_shift	Causal flow v4	0.773 +/- 0.006	0.628	0.242	0.534	0.113	0.555
combined_hard_shift	Causal flow v5	0.760 +/- 0.006	0.595	0.271	0.543	0.098	0.533
combined_hard_shift	Oracle	1.000 +/- 0.000	0.986	0.000	0.000	0.011	0.966
distractor_contact_shift	Magnitude	0.428 +/- 0.003	0.155	0.995	0.716	0.316	-0.047
distractor_contact_shift	Action projection	0.616 +/- 0.014	0.442	0.539	0.412	0.184	0.335
distractor_contact_shift	Rigid cluster	0.550 +/- 0.003	0.275	0.889	0.683	0.255	0.115
distractor_contact_shift	Temporal difference	0.427 +/- 0.004	0.150	0.972	0.739	0.316	-0.047
distractor_contact_shift	Noncausal proxy	0.676 +/- 0.004	0.498	0.459	0.528	0.200	0.376
distractor_contact_shift	Learned correlation	0.773 +/- 0.008	0.637	0.308	0.342	0.156	0.546
distractor_contact_shift	Calibrated tree	0.787 +/- 0.008	0.636	0.282	0.352	0.123	0.564
distractor_contact_shift	Contrastive action	0.789 +/- 0.009	0.678	0.250	0.280	0.127	0.599
distractor_contact_shift	Scene-flow transformer	0.723 +/- 0.008	0.561	0.413	0.452	0.170	0.459
distractor_contact_shift	World model	0.791 +/- 0.008	0.667	0.254	0.428	0.083	0.602
distractor_contact_shift	Causal flow v4	0.788 +/- 0.009	0.666	0.267	0.454	0.098	0.593
distractor_contact_shift	Causal flow v5	0.777 +/- 0.008	0.677	0.292	0.475	0.097	0.598
distractor_contact_shift	Oracle	1.000 +/- 0.000	0.978	0.000	0.000	0.014	0.960
ego_motion_shift	Magnitude	0.549 +/- 0.003	0.381	0.678	0.652	0.197	0.233
ego_motion_shift	Action projection	0.653 +/- 0.015	0.548	0.427	0.417	0.144	0.451
ego_motion_shift	Rigid cluster	0.679 +/- 0.003	0.511	0.570	0.549	0.169	0.392
ego_motion_shift	Temporal difference	0.538 +/- 0.004	0.328	0.789	0.683	0.222	0.170
ego_motion_shift	Noncausal proxy	0.837 +/- 0.004	0.695	0.214	0.311	0.091	0.634
ego_motion_shift	Learned correlation	0.864 +/- 0.005	0.736	0.185	0.260	0.100	0.677
ego_motion_shift	Calibrated tree	0.869 +/- 0.004	0.742	0.180	0.265	0.058	0.700
ego_motion_shift	Contrastive action	0.895 +/- 0.004	0.811	0.123	0.195	0.048	0.766
ego_motion_shift	Scene-flow transformer	0.832 +/- 0.006	0.713	0.248	0.307	0.081	0.652
ego_motion_shift	World model	0.872 +/- 0.004	0.772	0.168	0.335	0.058	0.721
ego_motion_shift	Causal flow v4	0.865 +/- 0.004	0.745	0.186	0.357	0.055	0.701
ego_motion_shift	Causal flow v5	0.850 +/- 0.005	0.766	0.215	0.376	0.070	0.705
ego_motion_shift	Oracle	1.000 +/- 0.000	0.973	0.000	0.000	0.002	0.961
occluded_interaction	Magnitude	0.418 +/- 0.004	0.228	0.666	0.740	0.223	0.078
occluded_interaction	Action projection	0.661 +/- 0.013	0.517	0.434	0.495	0.145	0.415
occluded_interaction	Rigid cluster	0.692 +/- 0.004	0.502	0.529	0.644	0.189	0.378
occluded_interaction	Temporal difference	0.442 +/- 0.005	0.241	0.791	0.758	0.259	0.063
occluded_interaction	Noncausal proxy	0.823 +/- 0.002	0.697	0.199	0.464	0.098	0.618
occluded_interaction	Learned correlation	0.855 +/- 0.005	0.689	0.177	0.435	0.100	0.629
occluded_interaction	Calibrated tree	0.858 +/- 0.004	0.741	0.182	0.448	0.072	0.676
occluded_interaction	Contrastive action	0.860 +/- 0.005	0.756	0.113	0.358	0.067	0.700
occluded_interaction	Scene-flow transformer	0.834 +/- 0.004	0.702	0.222	0.465	0.094	0.630
occluded_interaction	World model	0.848 +/- 0.005	0.730	0.169	0.460	0.052	0.676
occluded_interaction	Causal flow v4	0.843 +/- 0.004	0.714	0.195	0.489	0.078	0.653
occluded_interaction	Causal flow v5	0.839 +/- 0.004	0.738	0.212	0.503	0.055	0.677
occluded_interaction	Oracle	1.000 +/- 0.000	0.981	0.000	0.000	0.009	0.964
passive_conveyor	Magnitude	0.603 +/- 0.004	0.373	0.677	0.759	0.228	0.222

Table 7: All per-split main metrics. (continued)

Split	Method	F1	Success	Passive false	Effect err.	Unsafe	Utility
passive_conveyor	Action projection	0.741 +/- 0.012	0.639	0.429	0.369	0.134	0.536
passive_conveyor	Rigid cluster	0.585 +/- 0.004	0.306	0.895	0.735	0.228	0.154
passive_conveyor	Temporal difference	0.584 +/- 0.005	0.327	0.765	0.748	0.231	0.174
passive_conveyor	Noncausal proxy	0.730 +/- 0.004	0.533	0.438	0.543	0.147	0.436
passive_conveyor	Learned correlation	0.858 +/- 0.007	0.766	0.209	0.233	0.078	0.703
passive_conveyor	Calibrated tree	0.880 +/- 0.005	0.787	0.165	0.221	0.064	0.736
passive_conveyor	Contrastive action	0.898 +/- 0.006	0.777	0.127	0.182	0.064	0.738
passive_conveyor	Scene-flow transformer	0.818 +/- 0.006	0.662	0.296	0.286	0.100	0.602
passive_conveyor	World model	0.885 +/- 0.005	0.802	0.152	0.308	0.059	0.747
passive_conveyor	Causal flow v4	0.872 +/- 0.006	0.741	0.175	0.330	0.056	0.700
passive_conveyor	Causal flow v5	0.865 +/- 0.006	0.761	0.193	0.345	0.062	0.711
passive_conveyor	Oracle	1.000 +/- 0.000	0.972	0.000	0.000	0.003	0.959
tool_slip_hidden_contact	Magnitude	0.398 +/- 0.004	0.237	0.792	0.726	0.272	0.045
tool_slip_hidden_contact	Action projection	0.619 +/- 0.010	0.477	0.469	0.370	0.178	0.369
tool_slip_hidden_contact	Rigid cluster	0.608 +/- 0.003	0.388	0.682	0.618	0.242	0.240
tool_slip_hidden_contact	Temporal difference	0.407 +/- 0.005	0.183	0.877	0.729	0.303	-0.013
tool_slip_hidden_contact	Noncausal proxy	0.694 +/- 0.004	0.594	0.263	0.320	0.117	0.511
tool_slip_hidden_contact	Learned correlation	0.777 +/- 0.005	0.716	0.197	0.222	0.116	0.633
tool_slip_hidden_contact	Calibrated tree	0.792 +/- 0.005	0.688	0.190	0.238	0.103	0.622
tool_slip_hidden_contact	Contrastive action	0.776 +/- 0.002	0.683	0.133	0.174	0.114	0.620
tool_slip_hidden_contact	Scene-flow transformer	0.775 +/- 0.005	0.653	0.257	0.294	0.141	0.570
tool_slip_hidden_contact	World model	0.788 +/- 0.006	0.700	0.168	0.322	0.095	0.632
tool_slip_hidden_contact	Causal flow v4	0.789 +/- 0.006	0.691	0.186	0.352	0.100	0.623
tool_slip_hidden_contact	Causal flow v5	0.788 +/- 0.006	0.686	0.207	0.373	0.092	0.621
tool_slip_hidden_contact	Oracle	1.000 +/- 0.000	0.986	0.000	0.000	0.014	0.965
transparent_depth_noise	Magnitude	0.423 +/- 0.005	0.250	0.756	0.742	0.239	0.080
transparent_depth_noise	Action projection	0.565 +/- 0.005	0.475	0.446	0.430	0.155	0.369
transparent_depth_noise	Rigid cluster	0.599 +/- 0.003	0.383	0.626	0.672	0.206	0.256
transparent_depth_noise	Temporal difference	0.419 +/- 0.004	0.192	0.824	0.788	0.275	0.017
transparent_depth_noise	Noncausal proxy	0.767 +/- 0.004	0.595	0.293	0.481	0.117	0.517
transparent_depth_noise	Learned correlation	0.829 +/- 0.003	0.717	0.192	0.306	0.106	0.647
transparent_depth_noise	Calibrated tree	0.838 +/- 0.003	0.747	0.189	0.329	0.089	0.676
transparent_depth_noise	Contrastive action	0.834 +/- 0.002	0.731	0.123	0.224	0.084	0.678
transparent_depth_noise	Scene-flow transformer	0.789 +/- 0.004	0.647	0.263	0.368	0.117	0.575
transparent_depth_noise	World model	0.823 +/- 0.004	0.753	0.177	0.359	0.069	0.686
transparent_depth_noise	Causal flow v4	0.814 +/- 0.004	0.703	0.204	0.396	0.069	0.647
transparent_depth_noise	Causal flow v5	0.804 +/- 0.005	0.714	0.220	0.406	0.083	0.647
transparent_depth_noise	Oracle	1.000 +/- 0.000	0.975	0.000	0.000	0.011	0.959

B FULL HARD-AGGREGATE PAIRWISE TABLE

Table 8: All hard-aggregate paired seed differences for v5.

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Magnitude	mask_f1	0.34113	0.00194	0.33918	10/10
Magnitude	target_success	0.43730	0.01875	0.41856	10/10
Magnitude	causal_utility	0.54603	0.01433	0.53170	10/10
Magnitude	passive_false_attribution	-0.53174	0.00299	-0.53473	10/10
Magnitude	effect_error	-0.31116	0.00889	-0.32005	10/10
Magnitude	unsafe_contact	-0.17285	0.01443	-0.18728	10/10
Magnitude	risk_upper	-0.15340	0.00086	-0.15427	10/10
Magnitude	intervention_consistency	0.33860	0.00084	0.33776	10/10
Action projection	mask_f1	0.16898	0.00439	0.16459	10/10
Action projection	target_success	0.19609	0.01482	0.18127	10/10
Action projection	causal_utility	0.23661	0.01307	0.22354	10/10
Action projection	passive_false_attribution	-0.22796	0.00973	-0.23770	10/10
Action projection	effect_error	0.00434	0.00410	0.00024	2/10
Action projection	unsafe_contact	-0.08008	0.01331	-0.09339	10/10
Action projection	risk_upper	-0.06164	0.00287	-0.06452	10/10
Action projection	intervention_consistency	0.13655	0.00246	0.13409	10/10
Rigid cluster	mask_f1	0.19070	0.00137	0.18933	10/10
Rigid cluster	target_success	0.30957	0.01973	0.28984	10/10
Rigid cluster	causal_utility	0.38491	0.01750	0.36741	10/10
Rigid cluster	passive_false_attribution	-0.46383	0.00308	-0.46690	10/10
Rigid cluster	effect_error	-0.23196	0.00808	-0.24004	10/10
Rigid cluster	unsafe_contact	-0.13476	0.01244	-0.14720	10/10
Rigid cluster	risk_upper	-0.11638	0.00083	-0.11721	10/10
Rigid cluster	intervention_consistency	0.16341	0.00088	0.16253	10/10
Temporal difference	mask_f1	0.34162	0.00195	0.33967	10/10

Table 8: All hard-aggregate paired seed differences for v5. (continued)

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Temporal difference	target_success	0.46992	0.01793	0.45199	10/10
Temporal difference	causal_utility	0.58454	0.01548	0.56906	10/10
Temporal difference	passive_false_attribution	-0.60627	0.00275	-0.60902	10/10
Temporal difference	effect_error	-0.32656	0.00885	-0.33541	10/10
Temporal difference	unsafe_contact	-0.19141	0.01656	-0.20796	10/10
Temporal difference	risk_upper	-0.17630	0.00073	-0.17703	10/10
Temporal difference	intervention_consistency	0.33070	0.00102	0.32968	10/10
Noncausal proxy	mask_f1	0.07554	0.00162	0.07393	10/10
Noncausal proxy	target_success	0.12871	0.01992	0.10879	10/10
Noncausal proxy	causal_utility	0.16109	0.01743	0.14366	10/10
Noncausal proxy	passive_false_attribution	-0.12249	0.00260	-0.12509	10/10
Noncausal proxy	effect_error	-0.04716	0.00707	-0.05423	10/10
Noncausal proxy	unsafe_contact	-0.06270	0.01428	-0.07698	10/10
Noncausal proxy	risk_upper	-0.02252	0.00072	-0.02323	10/10
Noncausal proxy	intervention_consistency	0.15675	0.00086	0.15589	10/10
Learned correlation	mask_f1	-0.00409	0.00155	-0.00563	0/10
Learned correlation	target_success	0.01172	0.01669	-0.00497	7/10
Learned correlation	causal_utility	0.01973	0.01364	0.00610	8/10
Learned correlation	passive_false_attribution	0.00668	0.00271	0.00397	0/10
Learned correlation	effect_error	0.11560	0.00382	0.11179	0/10
Learned correlation	unsafe_contact	-0.03144	0.00986	-0.04130	10/10
Learned correlation	risk_upper	0.01740	0.00077	0.01663	0/10
Learned correlation	intervention_consistency	0.07168	0.00047	0.07121	10/10
Calibrated tree	mask_f1	-0.01578	0.00068	-0.01647	0/10
Calibrated tree	target_success	-0.01055	0.02537	-0.03591	5/10
Calibrated tree	causal_utility	-0.01053	0.01917	-0.02970	5/10
Calibrated tree	passive_false_attribution	0.02189	0.00157	0.02033	0/10
Calibrated tree	effect_error	0.10800	0.00393	0.10406	0/10
Calibrated tree	unsafe_contact	-0.00625	0.01008	-0.01633	6/10
Calibrated tree	risk_upper	-0.00485	0.00043	-0.00528	10/10
Calibrated tree	intervention_consistency	0.05197	0.00028	0.05169	10/10
Contrastive action	mask_f1	-0.02076	0.00152	-0.02228	0/10
Contrastive action	target_success	-0.02637	0.01243	-0.03880	0/10
Contrastive action	causal_utility	-0.02955	0.00914	-0.03869	0/10
Contrastive action	passive_false_attribution	0.07392	0.00181	0.07210	0/10
Contrastive action	effect_error	0.17970	0.00508	0.17462	0/10
Contrastive action	unsafe_contact	-0.01035	0.00927	-0.01962	6/10
Contrastive action	risk_upper	0.04047	0.00053	0.03994	0/10
Contrastive action	intervention_consistency	0.06792	0.00061	0.06731	10/10
Scene-flow transformer	mask_f1	0.02471	0.00169	0.02302	10/10
Scene-flow transformer	target_success	0.04981	0.02255	0.02726	9/10
Scene-flow transformer	causal_utility	0.06707	0.01873	0.04834	10/10
Scene-flow transformer	passive_false_attribution	-0.06439	0.00354	-0.06793	10/10
Scene-flow transformer	effect_error	0.05428	0.00409	0.05018	0/10
Scene-flow transformer	unsafe_contact	-0.04219	0.01334	-0.05552	10/10
Scene-flow transformer	risk_upper	-0.00393	0.00098	-0.00491	10/10
Scene-flow transformer	intervention_consistency	0.07161	0.00054	0.07107	10/10
World model	mask_f1	-0.01470	0.00068	-0.01538	0/10
World model	target_success	-0.02246	0.01941	-0.04187	0/10
World model	causal_utility	-0.02280	0.01375	-0.03655	0/10
World model	passive_false_attribution	0.04169	0.00095	0.04074	0/10
World model	effect_error	0.04314	0.00156	0.04158	0/10
World model	unsafe_contact	0.00469	0.00931	-0.00462	4/10
World model	risk_upper	0.01239	0.00023	0.01215	0/10
World model	intervention_consistency	0.03792	0.00029	0.03763	10/10
Causal flow v4	mask_f1	-0.00859	0.00068	-0.00927	0/10
Causal flow v4	target_success	0.00117	0.01702	-0.01585	5/10
Causal flow v4	causal_utility	-0.00061	0.01524	-0.01585	5/10
Causal flow v4	passive_false_attribution	0.02167	0.00057	0.02110	0/10
Causal flow v4	effect_error	0.01552	0.00062	0.01490	0/10
Causal flow v4	unsafe_contact	-0.00293	0.01008	-0.01301	6/10
Causal flow v4	risk_upper	0.03391	0.00015	0.03376	0/10
Causal flow v4	intervention_consistency	0.02019	0.00028	0.01991	10/10
Oracle	mask_f1	-0.18301	0.00175	-0.18477	0/10
Oracle	target_success	-0.27031	0.01708	-0.28739	0/10
Oracle	causal_utility	-0.31411	0.01444	-0.32855	0/10
Oracle	passive_false_attribution	0.22730	0.00267	0.22463	0/10
Oracle	effect_error	0.41542	0.00456	0.41086	0/10
Oracle	unsafe_contact	0.06777	0.00946	0.05831	0/10
Oracle	risk_upper	0.16681	0.00074	0.16607	0/10
Oracle	intervention_consistency	-0.15839	0.00098	-0.15937	0/10

C FULL STRESS SWEEP

Table 9: All stress axes and levels.

Axis	Level	Method	F1	Success	Unsafe	Utility
passive_flow	0.00	Action projection	0.59618	0.54688	0.17813	0.41648
passive_flow	0.00	Learned correlation	0.78031	0.66250	0.13437	0.57641
passive_flow	0.00	Calibrated tree	0.79256	0.67500	0.10625	0.59963
passive_flow	0.00	Contrastive action	0.77945	0.70937	0.11563	0.62570
passive_flow	0.00	World model	0.79616	0.68437	0.12812	0.60178
passive_flow	0.00	Causal flow v5	0.78662	0.64375	0.06250	0.59271
passive_flow	0.00	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	0.25	Action projection	0.57593	0.47813	0.18438	0.35260
passive_flow	0.25	Learned correlation	0.77416	0.66563	0.13750	0.57545
passive_flow	0.25	Calibrated tree	0.78488	0.66875	0.11875	0.58824
passive_flow	0.25	Contrastive action	0.76979	0.72500	0.11875	0.63231
passive_flow	0.25	World model	0.78753	0.67812	0.12812	0.59350
passive_flow	0.25	Causal flow v5	0.78024	0.65000	0.05937	0.59538
passive_flow	0.25	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	0.50	Action projection	0.58215	0.51562	0.19687	0.37646
passive_flow	0.50	Learned correlation	0.77714	0.66875	0.13750	0.57840
passive_flow	0.50	Calibrated tree	0.79508	0.66875	0.11250	0.59473
passive_flow	0.50	Contrastive action	0.78258	0.70937	0.10938	0.62958
passive_flow	0.50	World model	0.79400	0.64687	0.13125	0.57132
passive_flow	0.50	Causal flow v5	0.78401	0.63125	0.07187	0.57761
passive_flow	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	0.75	Action projection	0.55039	0.45625	0.21563	0.30797
passive_flow	0.75	Learned correlation	0.73344	0.60000	0.15000	0.50418
passive_flow	0.75	Calibrated tree	0.76422	0.60938	0.14062	0.52964
passive_flow	0.75	Contrastive action	0.73390	0.65625	0.13125	0.56290
passive_flow	0.75	World model	0.77151	0.64687	0.13125	0.55955
passive_flow	0.75	Causal flow v5	0.76416	0.58437	0.07500	0.53221
passive_flow	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	1.00	Action projection	0.56772	0.45000	0.20000	0.31476
passive_flow	1.00	Learned correlation	0.71883	0.57500	0.15937	0.47291
passive_flow	1.00	Calibrated tree	0.74573	0.60313	0.14062	0.51467
passive_flow	1.00	Contrastive action	0.72130	0.63438	0.13750	0.53488
passive_flow	1.00	World model	0.75711	0.64687	0.14375	0.54905
passive_flow	1.00	Causal flow v5	0.74114	0.56250	0.07187	0.50934
passive_flow	1.00	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	1.25	Action projection	0.55537	0.42500	0.22187	0.27988
passive_flow	1.25	Learned correlation	0.70623	0.53125	0.16875	0.42796
passive_flow	1.25	Calibrated tree	0.72919	0.59062	0.14688	0.49188
passive_flow	1.25	Contrastive action	0.70196	0.58437	0.15625	0.47943
passive_flow	1.25	World model	0.73555	0.59375	0.13125	0.50633
passive_flow	1.25	Causal flow v5	0.72297	0.55312	0.07187	0.49332
passive_flow	1.25	Oracle	1.00000	0.97500	0.01875	0.95575
passive_flow	1.50	Action projection	0.53823	0.41250	0.20938	0.26904
passive_flow	1.50	Learned correlation	0.70021	0.53438	0.20312	0.41220
passive_flow	1.50	Calibrated tree	0.72093	0.56250	0.16250	0.46014
passive_flow	1.50	Contrastive action	0.69739	0.57812	0.14688	0.47216
passive_flow	1.50	World model	0.71907	0.58125	0.13437	0.48894
passive_flow	1.50	Causal flow v5	0.70682	0.50625	0.07187	0.45322
passive_flow	1.50	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	0.00	Action projection	0.62432	0.52500	0.17813	0.40176
occlusion	0.00	Learned correlation	0.75733	0.62813	0.14375	0.54088
occlusion	0.00	Calibrated tree	0.77829	0.66563	0.14062	0.58070
occlusion	0.00	Contrastive action	0.77837	0.70312	0.12187	0.61706
occlusion	0.00	World model	0.79798	0.70625	0.13437	0.61543
occlusion	0.00	Causal flow v5	0.77839	0.65312	0.06875	0.59473
occlusion	0.00	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	0.25	Action projection	0.61074	0.52812	0.19687	0.39546
occlusion	0.25	Learned correlation	0.75518	0.64375	0.14375	0.55041
occlusion	0.25	Calibrated tree	0.77301	0.68437	0.14062	0.59017
occlusion	0.25	Contrastive action	0.76288	0.68437	0.11875	0.59973
occlusion	0.25	World model	0.78442	0.68125	0.11875	0.59608
occlusion	0.25	Causal flow v5	0.76616	0.60938	0.07187	0.55624
occlusion	0.25	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	0.50	Action projection	0.56633	0.47187	0.19687	0.33554
occlusion	0.50	Learned correlation	0.73827	0.60625	0.16875	0.50509
occlusion	0.50	Calibrated tree	0.76651	0.65000	0.13125	0.56460
occlusion	0.50	Contrastive action	0.74215	0.68125	0.11563	0.58967
occlusion	0.50	World model	0.77771	0.66563	0.13125	0.57788
occlusion	0.50	Causal flow v5	0.76561	0.60000	0.07812	0.54555
occlusion	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	0.75	Action projection	0.54362	0.41875	0.22187	0.28128
occlusion	0.75	Learned correlation	0.72915	0.58125	0.15313	0.48701
occlusion	0.75	Calibrated tree	0.75770	0.61562	0.13125	0.53354
occlusion	0.75	Contrastive action	0.72793	0.61875	0.13750	0.53178

Table 9: All stress axes and levels. (continued)

Axis	Level	Method	F1	Success	Unsafe	Utility
occlusion	0.75	World model	0.76174	0.63125	0.13437	0.54517
occlusion	0.75	Causal flow v5	0.75480	0.60938	0.06875	0.54933
occlusion	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	1.00	Action projection	0.52931	0.40000	0.21563	0.26429
occlusion	1.00	Learned correlation	0.73571	0.59688	0.14375	0.50317
occlusion	1.00	Calibrated tree	0.76457	0.63125	0.12812	0.54557
occlusion	1.00	Contrastive action	0.72378	0.66250	0.13125	0.56491
occlusion	1.00	World model	0.75487	0.60938	0.13437	0.52375
occlusion	1.00	Causal flow v5	0.74835	0.58437	0.08125	0.52107
occlusion	1.00	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	1.25	Action projection	0.51132	0.41250	0.19687	0.27517
occlusion	1.25	Learned correlation	0.73656	0.60625	0.16562	0.50004
occlusion	1.25	Calibrated tree	0.76411	0.60625	0.12812	0.52395
occlusion	1.25	Contrastive action	0.69954	0.58437	0.14688	0.49732
occlusion	1.25	World model	0.74759	0.59375	0.15313	0.50111
occlusion	1.25	Causal flow v5	0.74300	0.54688	0.08750	0.48800
occlusion	1.25	Oracle	1.00000	0.97500	0.01875	0.95575
occlusion	1.50	Action projection	0.47510	0.36562	0.22500	0.22005
occlusion	1.50	Learned correlation	0.72597	0.55937	0.14375	0.47101
occlusion	1.50	Calibrated tree	0.75474	0.60625	0.14062	0.51506
occlusion	1.50	Contrastive action	0.67402	0.57812	0.13750	0.48814
occlusion	1.50	World model	0.73855	0.60625	0.14062	0.50884
occlusion	1.50	Causal flow v5	0.73688	0.56250	0.07187	0.50123
occlusion	1.50	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	0.00	Action projection	0.58712	0.48750	0.18438	0.36038
articulation_delay	0.00	Learned correlation	0.74619	0.61875	0.15000	0.52290
articulation_delay	0.00	Calibrated tree	0.76969	0.64062	0.12500	0.55914
articulation_delay	0.00	Contrastive action	0.75012	0.65938	0.14062	0.56747
articulation_delay	0.00	World model	0.77402	0.66250	0.12187	0.57642
articulation_delay	0.00	Causal flow v5	0.76369	0.59688	0.06875	0.54462
articulation_delay	0.00	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	0.25	Action projection	0.57538	0.49375	0.18750	0.36013
articulation_delay	0.25	Learned correlation	0.73342	0.60313	0.14375	0.51103
articulation_delay	0.25	Calibrated tree	0.75669	0.63750	0.13125	0.55098
articulation_delay	0.25	Contrastive action	0.73414	0.65000	0.13750	0.55750
articulation_delay	0.25	World model	0.75925	0.62500	0.13437	0.54052
articulation_delay	0.25	Causal flow v5	0.75309	0.59688	0.07187	0.54038
articulation_delay	0.25	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	0.50	Action projection	0.55498	0.42188	0.19062	0.29704
articulation_delay	0.50	Learned correlation	0.72978	0.60938	0.16875	0.50374
articulation_delay	0.50	Calibrated tree	0.75663	0.63750	0.13750	0.54832
articulation_delay	0.50	Contrastive action	0.72840	0.65312	0.11250	0.56570
articulation_delay	0.50	World model	0.76421	0.63750	0.13125	0.55173
articulation_delay	0.50	Causal flow v5	0.75421	0.60000	0.06875	0.54348
articulation_delay	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	0.75	Action projection	0.56611	0.47813	0.22187	0.33074
articulation_delay	0.75	Learned correlation	0.72773	0.61250	0.16250	0.50724
articulation_delay	0.75	Calibrated tree	0.75627	0.63125	0.13437	0.54504
articulation_delay	0.75	Contrastive action	0.72668	0.64062	0.13125	0.54984
articulation_delay	0.75	World model	0.76163	0.64062	0.13125	0.55210
articulation_delay	0.75	Causal flow v5	0.75349	0.60000	0.08125	0.53756
articulation_delay	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	1.00	Action projection	0.55758	0.43438	0.20938	0.29824
articulation_delay	1.00	Learned correlation	0.72430	0.59375	0.15313	0.49696
articulation_delay	1.00	Calibrated tree	0.75405	0.62813	0.12500	0.54602
articulation_delay	1.00	Contrastive action	0.72341	0.65000	0.13125	0.55615
articulation_delay	1.00	World model	0.76695	0.64375	0.13437	0.55566
articulation_delay	1.00	Causal flow v5	0.75840	0.58125	0.07812	0.52823
articulation_delay	1.00	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	1.25	Action projection	0.56096	0.45000	0.21563	0.31086
articulation_delay	1.25	Learned correlation	0.73951	0.60000	0.14062	0.51251
articulation_delay	1.25	Calibrated tree	0.76733	0.64687	0.12812	0.56348
articulation_delay	1.25	Contrastive action	0.73969	0.66250	0.13125	0.57155
articulation_delay	1.25	World model	0.76857	0.64375	0.11875	0.56157
articulation_delay	1.25	Causal flow v5	0.76178	0.56563	0.06875	0.52189
articulation_delay	1.25	Oracle	1.00000	0.97500	0.01875	0.95575
articulation_delay	1.50	Action projection	0.54951	0.45312	0.22187	0.30467
articulation_delay	1.50	Learned correlation	0.72047	0.57812	0.16875	0.47627
articulation_delay	1.50	Calibrated tree	0.75395	0.61875	0.14062	0.53253
articulation_delay	1.50	Contrastive action	0.72093	0.60938	0.13750	0.52267
articulation_delay	1.50	World model	0.76035	0.64062	0.14375	0.54841
articulation_delay	1.50	Causal flow v5	0.75127	0.58125	0.07187	0.52815
articulation_delay	1.50	Oracle	1.00000	0.97500	0.01875	0.95575
distractor_contact	0.00	Action projection	0.61883	0.55000	0.13750	0.44041
distractor_contact	0.00	Learned correlation	0.81613	0.69375	0.11250	0.62016
distractor_contact	0.00	Calibrated tree	0.84643	0.70000	0.08125	0.65138
distractor_contact	0.00	Contrastive action	0.82844	0.75000	0.07812	0.68787

Table 9: All stress axes and levels. (continued)

Axis	Level	Method	F1	Success	Unsafe	Utility
distractor_contact	0.00	World model	0.83070	0.73438	0.10625	0.65894
distractor_contact	0.00	Causal flow v5	0.81597	0.66563	0.05625	0.62233
distractor_contact	0.00	Oracle	1.00000	0.97500	0.01250	0.95800
distractor_contact	0.25	Action projection	0.58631	0.48750	0.17188	0.36483
distractor_contact	0.25	Learned correlation	0.77698	0.65000	0.13125	0.56477
distractor_contact	0.25	Calibrated tree	0.80441	0.65938	0.09688	0.59808
distractor_contact	0.25	Contrastive action	0.78449	0.71250	0.09062	0.63806
distractor_contact	0.25	World model	0.80566	0.70312	0.11250	0.62238
distractor_contact	0.25	Causal flow v5	0.79771	0.65000	0.05625	0.60229
distractor_contact	0.25	Oracle	1.00000	0.97500	0.01875	0.95575
distractor_contact	0.50	Action projection	0.56575	0.46875	0.21250	0.32396
distractor_contact	0.50	Learned correlation	0.73550	0.58750	0.18125	0.48347
distractor_contact	0.50	Calibrated tree	0.76395	0.64062	0.12500	0.55601
distractor_contact	0.50	Contrastive action	0.73971	0.63125	0.13125	0.54577
distractor_contact	0.50	World model	0.77084	0.65312	0.12812	0.56652
distractor_contact	0.50	Causal flow v5	0.75929	0.61250	0.07187	0.55323
distractor_contact	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
distractor_contact	0.75	Action projection	0.57226	0.47500	0.18438	0.34688
distractor_contact	0.75	Learned correlation	0.72105	0.55937	0.15625	0.46612
distractor_contact	0.75	Calibrated tree	0.74631	0.61562	0.12812	0.52866
distractor_contact	0.75	Contrastive action	0.72228	0.63125	0.12812	0.53822
distractor_contact	0.75	World model	0.74783	0.61250	0.14375	0.52054
distractor_contact	0.75	Causal flow v5	0.73527	0.56250	0.08125	0.50261
distractor_contact	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
distractor_contact	1.00	Action projection	0.55751	0.45625	0.21875	0.30828
distractor_contact	1.00	Learned correlation	0.70388	0.54375	0.16875	0.43895
distractor_contact	1.00	Calibrated tree	0.72772	0.57812	0.14688	0.48368
distractor_contact	1.00	Contrastive action	0.70247	0.60000	0.13750	0.49938
distractor_contact	1.00	World model	0.73211	0.58125	0.15313	0.48558
distractor_contact	1.00	Causal flow v5	0.72176	0.50625	0.09688	0.44890
distractor_contact	1.00	Oracle	1.00000	0.97500	0.02187	0.95463
distractor_contact	1.25	Action projection	0.53632	0.39687	0.24063	0.24871
distractor_contact	1.25	Learned correlation	0.67954	0.51875	0.20625	0.39546
distractor_contact	1.25	Calibrated tree	0.69773	0.55000	0.17813	0.43733
distractor_contact	1.25	Contrastive action	0.67441	0.54063	0.17500	0.42954
distractor_contact	1.25	World model	0.69418	0.55000	0.17500	0.43477
distractor_contact	1.25	Causal flow v5	0.68714	0.48125	0.10938	0.40667
distractor_contact	1.25	Oracle	1.00000	0.97500	0.02500	0.95350
distractor_contact	1.50	Action projection	0.53306	0.40625	0.26562	0.23877
distractor_contact	1.50	Learned correlation	0.65616	0.46875	0.24063	0.32906
distractor_contact	1.50	Calibrated tree	0.67492	0.49688	0.20000	0.37392
distractor_contact	1.50	Contrastive action	0.64604	0.49062	0.18438	0.36890
distractor_contact	1.50	World model	0.68080	0.50313	0.18438	0.38724
distractor_contact	1.50	Causal flow v5	0.67568	0.45625	0.10938	0.37932
distractor_contact	1.50	Oracle	1.00000	0.97500	0.02500	0.95350
ego_motion	0.00	Action projection	0.58711	0.48750	0.21250	0.34263
ego_motion	0.00	Learned correlation	0.74066	0.59375	0.16562	0.49636
ego_motion	0.00	Calibrated tree	0.77261	0.62813	0.12500	0.55149
ego_motion	0.00	Contrastive action	0.74484	0.64687	0.12500	0.56181
ego_motion	0.00	World model	0.77866	0.66563	0.12812	0.57888
ego_motion	0.00	Causal flow v5	0.77148	0.62813	0.06250	0.57355
ego_motion	0.00	Oracle	1.00000	0.97500	0.01875	0.95575
ego_motion	0.25	Action projection	0.58051	0.47187	0.19687	0.34041
ego_motion	0.25	Learned correlation	0.74405	0.60938	0.15313	0.51453
ego_motion	0.25	Calibrated tree	0.77034	0.64375	0.12500	0.56220
ego_motion	0.25	Contrastive action	0.74475	0.64687	0.13437	0.55953
ego_motion	0.25	World model	0.77309	0.64062	0.13437	0.55576
ego_motion	0.25	Causal flow v5	0.76514	0.61875	0.06563	0.56132
ego_motion	0.25	Oracle	1.00000	0.97500	0.01875	0.95575
ego_motion	0.50	Action projection	0.53655	0.45625	0.20312	0.31427
ego_motion	0.50	Learned correlation	0.73029	0.57812	0.16562	0.48229
ego_motion	0.50	Calibrated tree	0.75933	0.63438	0.13125	0.54917
ego_motion	0.50	Contrastive action	0.73118	0.64687	0.13125	0.55579
ego_motion	0.50	World model	0.76662	0.65000	0.13125	0.56183
ego_motion	0.50	Causal flow v5	0.75526	0.60000	0.06875	0.54411
ego_motion	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
ego_motion	0.75	Action projection	0.54195	0.44688	0.20625	0.31247
ego_motion	0.75	Learned correlation	0.73799	0.58437	0.16562	0.48990
ego_motion	0.75	Calibrated tree	0.76289	0.65000	0.14062	0.55782
ego_motion	0.75	Contrastive action	0.73741	0.64687	0.12187	0.56084
ego_motion	0.75	World model	0.76493	0.63438	0.12187	0.55296
ego_motion	0.75	Causal flow v5	0.75329	0.60000	0.07812	0.53994
ego_motion	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
ego_motion	1.00	Action projection	0.51845	0.43438	0.21875	0.29588
ego_motion	1.00	Learned correlation	0.73855	0.59688	0.16250	0.50084
ego_motion	1.00	Calibrated tree	0.76286	0.62187	0.12500	0.54443
ego_motion	1.00	Contrastive action	0.73995	0.66875	0.13437	0.57262

Table 9: All stress axes and levels. (continued)

Axis	Level	Method	F1	Success	Unsafe	Utility
ego.motion	1.00	World model	0.76053	0.64062	0.13750	0.54952
ego.motion	1.00	Causal flow v5	0.74497	0.58750	0.07500	0.52894
ego.motion	1.00	Oracle	1.00000	0.97500	0.01875	0.95575
ego.motion	1.25	Action projection	0.52328	0.44375	0.21250	0.30566
ego.motion	1.25	Learned correlation	0.73436	0.58437	0.14688	0.49627
ego.motion	1.25	Calibrated tree	0.75965	0.62187	0.11250	0.54683
ego.motion	1.25	Contrastive action	0.73614	0.65312	0.12812	0.56332
ego.motion	1.25	World model	0.75362	0.61562	0.12812	0.52870
ego.motion	1.25	Causal flow v5	0.73907	0.58750	0.08125	0.52009
ego.motion	1.25	Oracle	1.00000	0.97500	0.01875	0.95575
ego.motion	1.50	Action projection	0.50361	0.41250	0.22500	0.27234
ego.motion	1.50	Learned correlation	0.72701	0.55937	0.17188	0.46689
ego.motion	1.50	Calibrated tree	0.74967	0.60313	0.12812	0.52361
ego.motion	1.50	Contrastive action	0.72671	0.63750	0.13750	0.54427
ego.motion	1.50	World model	0.74047	0.57188	0.13125	0.49089
ego.motion	1.50	Causal flow v5	0.73002	0.55937	0.08438	0.49403
ego.motion	1.50	Oracle	1.00000	0.97500	0.01875	0.95575
combined	0.00	Action projection	0.87932	0.82812	0.08750	0.75599
combined	0.00	Learned correlation	0.91433	0.84062	0.06875	0.78918
combined	0.00	Calibrated tree	0.92636	0.87500	0.04688	0.82812
combined	0.00	Contrastive action	0.95408	0.92500	0.02813	0.88286
combined	0.00	World model	0.92266	0.85938	0.06875	0.80966
combined	0.00	Causal flow v5	0.88846	0.82188	0.04688	0.77966
combined	0.00	Oracle	1.00000	0.97500	0.01250	0.95800
combined	0.25	Action projection	0.70530	0.61250	0.15625	0.49984
combined	0.25	Learned correlation	0.85296	0.76562	0.09375	0.69732
combined	0.25	Calibrated tree	0.85252	0.75938	0.09375	0.69292
combined	0.25	Contrastive action	0.87646	0.80312	0.06875	0.74881
combined	0.25	World model	0.85749	0.77187	0.11563	0.69721
combined	0.25	Causal flow v5	0.84009	0.73750	0.05312	0.68935
combined	0.25	Oracle	1.00000	0.97500	0.01250	0.95800
combined	0.50	Action projection	0.57568	0.50938	0.20000	0.36568
combined	0.50	Learned correlation	0.77518	0.65312	0.13125	0.57103
combined	0.50	Calibrated tree	0.79117	0.69375	0.11875	0.61040
combined	0.50	Contrastive action	0.78384	0.72500	0.09062	0.65020
combined	0.50	World model	0.79454	0.69688	0.12812	0.60967
combined	0.50	Causal flow v5	0.78483	0.63438	0.07500	0.58006
combined	0.50	Oracle	1.00000	0.97500	0.01875	0.95575
combined	0.75	Action projection	0.52699	0.42500	0.21563	0.28823
combined	0.75	Learned correlation	0.71763	0.59375	0.16250	0.49187
combined	0.75	Calibrated tree	0.74017	0.60938	0.13437	0.52261
combined	0.75	Contrastive action	0.70622	0.62500	0.13750	0.53061
combined	0.75	World model	0.73497	0.61250	0.15000	0.51341
combined	0.75	Causal flow v5	0.72865	0.55000	0.08125	0.49066
combined	0.75	Oracle	1.00000	0.97500	0.01875	0.95575
combined	1.00	Action projection	0.46340	0.33437	0.23750	0.19169
combined	1.00	Learned correlation	0.66332	0.48438	0.19375	0.37766
combined	1.00	Calibrated tree	0.69082	0.56875	0.17188	0.45584
combined	1.00	Contrastive action	0.63600	0.52812	0.15625	0.42423
combined	1.00	World model	0.67097	0.51875	0.15625	0.41489
combined	1.00	Causal flow v5	0.66615	0.47813	0.09688	0.40383
combined	1.00	Oracle	1.00000	0.97500	0.02500	0.95350
combined	1.25	Action projection	0.40044	0.27187	0.27187	0.10265
combined	1.25	Learned correlation	0.59003	0.38750	0.23125	0.25527
combined	1.25	Calibrated tree	0.62720	0.45000	0.18125	0.33208
combined	1.25	Contrastive action	0.54196	0.39375	0.20312	0.26383
combined	1.25	World model	0.60644	0.44375	0.18438	0.31535
combined	1.25	Causal flow v5	0.60866	0.40937	0.11563	0.31359
combined	1.25	Oracle	1.00000	0.97500	0.02500	0.95350
combined	1.50	Action projection	0.37103	0.22187	0.28750	0.04445
combined	1.50	Learned correlation	0.55224	0.32500	0.26875	0.17170
combined	1.50	Calibrated tree	0.59287	0.38438	0.22187	0.24678
combined	1.50	Contrastive action	0.49532	0.31562	0.20625	0.17733
combined	1.50	World model	0.56339	0.37500	0.19062	0.23756
combined	1.50	Causal flow v5	0.56934	0.35000	0.12812	0.24002
combined	1.50	Oracle	1.00000	0.95312	0.02813	0.93706

D COMBINED STRESS SEED METRICS

Table 10: Seed-level combined stress metrics.

Level	Method	Seed	F1	Success	Unsafe	Utility
0.00	Action projection	0	0.85926	0.87500	0.15625	0.75477
0.00	Action projection	1	0.89059	0.78125	0.03125	0.74996
0.00	Action projection	2	0.84575	0.93750	0.06250	0.82384
0.00	Action projection	3	0.88491	0.81250	0.06250	0.75774
0.00	Action projection	4	0.87388	0.78125	0.12500	0.70893
0.00	Action projection	5	0.86384	0.71875	0.09375	0.67155
0.00	Action projection	6	0.91029	0.84375	0.06250	0.79175
0.00	Action projection	7	0.89657	0.90625	0.03125	0.83633
0.00	Action projection	8	0.87810	0.87500	0.12500	0.77378
0.00	Action projection	9	0.88997	0.75000	0.12500	0.69129
0.00	Calibrated tree	0	0.93427	0.90625	0.03125	0.85905
0.00	Calibrated tree	1	0.92082	0.81250	0.09375	0.76514
0.00	Calibrated tree	2	0.92252	0.90625	0.03125	0.85287
0.00	Calibrated tree	3	0.93798	0.90625	0.00000	0.87374
0.00	Calibrated tree	4	0.91969	0.87500	0.09375	0.80754
0.00	Calibrated tree	5	0.92768	0.78125	0.06250	0.75818
0.00	Calibrated tree	6	0.92424	0.87500	0.03125	0.83321
0.00	Calibrated tree	7	0.93298	0.87500	0.03125	0.83614
0.00	Calibrated tree	8	0.93527	0.93750	0.09375	0.85899
0.00	Calibrated tree	9	0.90813	0.87500	0.00000	0.83634
0.00	Causal flow v5	0	0.90429	0.87500	0.12500	0.79564
0.00	Causal flow v5	1	0.88162	0.84375	0.00000	0.80957
0.00	Causal flow v5	2	0.88017	0.81250	0.03125	0.77265
0.00	Causal flow v5	3	0.89502	0.84375	0.03125	0.80549
0.00	Causal flow v5	4	0.88182	0.84375	0.03125	0.79713
0.00	Causal flow v5	5	0.88753	0.81250	0.03125	0.77888
0.00	Causal flow v5	6	0.88879	0.78125	0.06250	0.74623
0.00	Causal flow v5	7	0.89322	0.71875	0.09375	0.69268
0.00	Causal flow v5	8	0.90793	0.87500	0.00000	0.84260
0.00	Causal flow v5	9	0.86420	0.81250	0.06250	0.75571
0.00	Contrastive action	0	0.95805	1.00000	0.03125	0.93547
0.00	Contrastive action	1	0.94751	0.93750	0.06250	0.87645
0.00	Contrastive action	2	0.94991	0.96875	0.00000	0.92116
0.00	Contrastive action	3	0.95987	0.84375	0.09375	0.80645
0.00	Contrastive action	4	0.94789	0.93750	0.03125	0.88745
0.00	Contrastive action	5	0.96163	0.90625	0.00000	0.88364
0.00	Contrastive action	6	0.95438	0.90625	0.00000	0.88066
0.00	Contrastive action	7	0.96244	0.93750	0.03125	0.89293
0.00	Contrastive action	8	0.95839	0.90625	0.03125	0.87061
0.00	Contrastive action	9	0.94071	0.90625	0.00000	0.87376
0.00	Learned correlation	0	0.91628	0.71875	0.03125	0.71825
0.00	Learned correlation	1	0.90464	0.87500	0.06250	0.81122
0.00	Learned correlation	2	0.90607	0.75000	0.06250	0.72347
0.00	Learned correlation	3	0.92864	0.81250	0.18750	0.73469
0.00	Learned correlation	4	0.90528	0.96875	0.06250	0.87591
0.00	Learned correlation	5	0.91450	0.75000	0.03125	0.74041
0.00	Learned correlation	6	0.91798	0.90625	0.06250	0.83955
0.00	Learned correlation	7	0.92020	0.87500	0.06250	0.81738
0.00	Learned correlation	8	0.92925	0.84375	0.03125	0.81151
0.00	Learned correlation	9	0.90042	0.90625	0.09375	0.81939
0.00	Oracle	0	1.00000	1.00000	0.03125	0.96875
0.00	Oracle	1	1.00000	1.00000	0.03125	0.96875
0.00	Oracle	2	1.00000	0.96875	0.03125	0.94687
0.00	Oracle	3	1.00000	1.00000	0.00000	0.98000
0.00	Oracle	4	1.00000	0.93750	0.00000	0.93625
0.00	Oracle	5	1.00000	0.96875	0.00000	0.95813
0.00	Oracle	6	1.00000	0.96875	0.00000	0.95813
0.00	Oracle	7	1.00000	0.96875	0.00000	0.95813
0.00	Oracle	8	1.00000	0.96875	0.00000	0.95813
0.00	Oracle	9	1.00000	0.96875	0.03125	0.94687
0.00	World model	0	0.93153	0.78125	0.00000	0.78313
0.00	World model	1	0.91772	0.84375	0.03125	0.81079
0.00	World model	2	0.92215	0.90625	0.09375	0.83160
0.00	World model	3	0.92964	0.84375	0.06250	0.80622
0.00	World model	4	0.91418	0.78125	0.00000	0.77552
0.00	World model	5	0.93053	0.90625	0.15625	0.81438
0.00	World model	6	0.91851	0.90625	0.09375	0.83215
0.00	World model	7	0.92572	0.93750	0.15625	0.83407
0.00	World model	8	0.93110	0.84375	0.09375	0.79349
0.00	World model	9	0.90557	0.84375	0.00000	0.81529
0.25	Action projection	0	0.70849	0.71875	0.15625	0.57266
0.25	Action projection	1	0.70849	0.53125	0.03125	0.48989
0.25	Action projection	2	0.68768	0.65625	0.12500	0.53102
0.25	Action projection	3	0.69481	0.65625	0.18750	0.51119
0.25	Action projection	4	0.70659	0.53125	0.18750	0.43024
0.25	Action projection	5	0.69174	0.56250	0.09375	0.48386
0.25	Action projection	6	0.69703	0.53125	0.31250	0.38524

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
0.25	Action projection	7	0.69020	0.59375	0.15625	0.48061
0.25	Action projection	8	0.73043	0.71875	0.21875	0.56691
0.25	Action projection	9	0.73757	0.62500	0.09375	0.54680
0.25	Calibrated tree	0	0.85783	0.78125	0.06250	0.72086
0.25	Calibrated tree	1	0.84815	0.68750	0.12500	0.62955
0.25	Calibrated tree	2	0.87722	0.87500	0.12500	0.77443
0.25	Calibrated tree	3	0.84882	0.71875	0.06250	0.67328
0.25	Calibrated tree	4	0.85767	0.81250	0.09375	0.73068
0.25	Calibrated tree	5	0.85261	0.59375	0.12500	0.56546
0.25	Calibrated tree	6	0.86592	0.90625	0.06250	0.81472
0.25	Calibrated tree	7	0.82788	0.68750	0.09375	0.63210
0.25	Calibrated tree	8	0.82394	0.78125	0.15625	0.67292
0.25	Calibrated tree	9	0.86520	0.75000	0.03125	0.71518
0.25	Causal flow v5	0	0.84958	0.78125	0.12500	0.69817
0.25	Causal flow v5	1	0.83588	0.65625	0.00000	0.65072
0.25	Causal flow v5	2	0.86607	0.81250	0.06250	0.74795
0.25	Causal flow v5	3	0.83010	0.71875	0.03125	0.67910
0.25	Causal flow v5	4	0.84196	0.78125	0.03125	0.72582
0.25	Causal flow v5	5	0.84121	0.68750	0.03125	0.66287
0.25	Causal flow v5	6	0.85350	0.71875	0.06250	0.67844
0.25	Causal flow v5	7	0.81774	0.65625	0.09375	0.60854
0.25	Causal flow v5	8	0.81350	0.78125	0.00000	0.72958
0.25	Causal flow v5	9	0.85136	0.78125	0.09375	0.71231
0.25	Contrastive action	0	0.86972	0.87500	0.12500	0.77492
0.25	Contrastive action	1	0.86862	0.75000	0.06250	0.71017
0.25	Contrastive action	2	0.89752	0.96875	0.09375	0.86808
0.25	Contrastive action	3	0.88265	0.81250	0.12500	0.73740
0.25	Contrastive action	4	0.87589	0.75000	0.09375	0.70184
0.25	Contrastive action	5	0.87876	0.75000	0.03125	0.72448
0.25	Contrastive action	6	0.88925	0.81250	0.03125	0.77642
0.25	Contrastive action	7	0.86475	0.78125	0.06250	0.73033
0.25	Contrastive action	8	0.85215	0.71875	0.06250	0.68062
0.25	Contrastive action	9	0.88526	0.81250	0.00000	0.78385
0.25	Learned correlation	0	0.85996	0.78125	0.03125	0.73249
0.25	Learned correlation	1	0.84399	0.68750	0.09375	0.63915
0.25	Learned correlation	2	0.87089	0.71875	0.12500	0.66216
0.25	Learned correlation	3	0.85690	0.68750	0.18750	0.61003
0.25	Learned correlation	4	0.85652	0.84375	0.09375	0.75249
0.25	Learned correlation	5	0.85701	0.68750	0.09375	0.64358
0.25	Learned correlation	6	0.85732	0.78125	0.06250	0.72374
0.25	Learned correlation	7	0.83471	0.81250	0.06250	0.73326
0.25	Learned correlation	8	0.82987	0.78125	0.03125	0.71977
0.25	Learned correlation	9	0.86246	0.87500	0.15625	0.75650
0.25	Oracle	0	1.00000	1.00000	0.03125	0.96875
0.25	Oracle	1	1.00000	1.00000	0.03125	0.96875
0.25	Oracle	2	1.00000	0.96875	0.03125	0.94687
0.25	Oracle	3	1.00000	1.00000	0.00000	0.98000
0.25	Oracle	4	1.00000	0.93750	0.00000	0.93625
0.25	Oracle	5	1.00000	0.96875	0.00000	0.95813
0.25	Oracle	6	1.00000	0.96875	0.00000	0.95813
0.25	Oracle	7	1.00000	0.96875	0.00000	0.95813
0.25	Oracle	8	1.00000	0.96875	0.00000	0.95813
0.25	Oracle	9	1.00000	0.96875	0.03125	0.94687
0.25	World model	0	0.86343	0.71875	0.06250	0.68015
0.25	World model	1	0.86014	0.84375	0.06250	0.76776
0.25	World model	2	0.88558	0.78125	0.12500	0.71352
0.25	World model	3	0.84970	0.68750	0.09375	0.64236
0.25	World model	4	0.85518	0.62500	0.06250	0.61050
0.25	World model	5	0.85586	0.84375	0.18750	0.72038
0.25	World model	6	0.87561	0.81250	0.15625	0.71902
0.25	World model	7	0.83633	0.81250	0.18750	0.69122
0.25	World model	8	0.82968	0.71875	0.18750	0.62276
0.25	World model	9	0.86344	0.87500	0.03125	0.80441
0.50	Action projection	0	0.56906	0.56250	0.18750	0.40670
0.50	Action projection	1	0.55486	0.37500	0.09375	0.30409
0.50	Action projection	2	0.60333	0.65625	0.09375	0.52486
0.50	Action projection	3	0.55459	0.53125	0.21875	0.35784
0.50	Action projection	4	0.58317	0.43750	0.28125	0.28249
0.50	Action projection	5	0.60009	0.50000	0.18750	0.38144
0.50	Action projection	6	0.59494	0.50000	0.28125	0.33489
0.50	Action projection	7	0.57259	0.46875	0.31250	0.29155
0.50	Action projection	8	0.54038	0.56250	0.18750	0.38435
0.50	Action projection	9	0.58383	0.50000	0.15625	0.38858
0.50	Calibrated tree	0	0.76115	0.68750	0.12500	0.59235
0.50	Calibrated tree	1	0.79286	0.62500	0.15625	0.55039
0.50	Calibrated tree	2	0.80626	0.78125	0.09375	0.68793
0.50	Calibrated tree	3	0.77279	0.62500	0.12500	0.55283

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
0.50	Calibrated tree	4	0.79271	0.68750	0.15625	0.59281
0.50	Calibrated tree	5	0.80910	0.65625	0.12500	0.58610
0.50	Calibrated tree	6	0.80123	0.87500	0.09375	0.75085
0.50	Calibrated tree	7	0.78035	0.59375	0.09375	0.54501
0.50	Calibrated tree	8	0.80200	0.78125	0.18750	0.65283
0.50	Calibrated tree	9	0.79327	0.62500	0.03125	0.59288
0.50	Causal flow v5	0	0.75366	0.68750	0.15625	0.57351
0.50	Causal flow v5	1	0.78617	0.50000	0.03125	0.50268
0.50	Causal flow v5	2	0.80472	0.68750	0.12500	0.61214
0.50	Causal flow v5	3	0.77125	0.56250	0.03125	0.53772
0.50	Causal flow v5	4	0.78349	0.68750	0.03125	0.62976
0.50	Causal flow v5	5	0.78822	0.65625	0.06250	0.60170
0.50	Causal flow v5	6	0.79461	0.59375	0.12500	0.53779
0.50	Causal flow v5	7	0.78823	0.56250	0.09375	0.52509
0.50	Causal flow v5	8	0.80858	0.81250	0.00000	0.74198
0.50	Causal flow v5	9	0.76937	0.59375	0.09375	0.53819
0.50	Contrastive action	0	0.76226	0.71875	0.12500	0.62476
0.50	Contrastive action	1	0.78614	0.71875	0.12500	0.63557
0.50	Contrastive action	2	0.80289	0.87500	0.12500	0.75067
0.50	Contrastive action	3	0.75680	0.62500	0.15625	0.54650
0.50	Contrastive action	4	0.77296	0.71875	0.12500	0.62958
0.50	Contrastive action	5	0.81047	0.71875	0.03125	0.67473
0.50	Contrastive action	6	0.79560	0.71875	0.06250	0.66172
0.50	Contrastive action	7	0.77750	0.71875	0.09375	0.64108
0.50	Contrastive action	8	0.78493	0.71875	0.06250	0.65887
0.50	Contrastive action	9	0.78885	0.71875	0.00000	0.67856
0.50	Learned correlation	0	0.75002	0.56250	0.09375	0.51213
0.50	Learned correlation	1	0.78425	0.71875	0.09375	0.63394
0.50	Learned correlation	2	0.79136	0.65625	0.15625	0.57206
0.50	Learned correlation	3	0.75610	0.53125	0.18750	0.45841
0.50	Learned correlation	4	0.76953	0.65625	0.12500	0.57256
0.50	Learned correlation	5	0.79904	0.68750	0.21875	0.57065
0.50	Learned correlation	6	0.79117	0.62500	0.06250	0.58360
0.50	Learned correlation	7	0.75313	0.75000	0.06250	0.65413
0.50	Learned correlation	8	0.76813	0.59375	0.12500	0.53076
0.50	Learned correlation	9	0.78902	0.75000	0.18750	0.62207
0.50	Oracle	0	1.00000	1.00000	0.06250	0.95750
0.50	Oracle	1	1.00000	1.00000	0.03125	0.96875
0.50	Oracle	2	1.00000	0.96875	0.03125	0.94687
0.50	Oracle	3	1.00000	1.00000	0.00000	0.98000
0.50	Oracle	4	1.00000	0.93750	0.00000	0.93625
0.50	Oracle	5	1.00000	0.96875	0.03125	0.94688
0.50	Oracle	6	1.00000	0.96875	0.00000	0.95813
0.50	Oracle	7	1.00000	0.96875	0.00000	0.95813
0.50	Oracle	8	1.00000	0.96875	0.00000	0.95813
0.50	Oracle	9	1.00000	0.96875	0.03125	0.94687
0.50	World model	0	0.76646	0.68750	0.09375	0.60213
0.50	World model	1	0.79988	0.81250	0.09375	0.70448
0.50	World model	2	0.82097	0.68750	0.18750	0.59491
0.50	World model	3	0.77186	0.62500	0.12500	0.55093
0.50	World model	4	0.78520	0.59375	0.06250	0.55517
0.50	World model	5	0.80004	0.71875	0.15625	0.61620
0.50	World model	6	0.80244	0.71875	0.18750	0.60666
0.50	World model	7	0.79201	0.71875	0.18750	0.60237
0.50	World model	8	0.81170	0.68750	0.15625	0.60375
0.50	World model	9	0.79482	0.71875	0.03125	0.66005
0.75	Action projection	0	0.54241	0.53125	0.21875	0.35883
0.75	Action projection	1	0.52765	0.31250	0.15625	0.22431
0.75	Action projection	2	0.51338	0.37500	0.12500	0.28497
0.75	Action projection	3	0.53551	0.43750	0.21875	0.30393
0.75	Action projection	4	0.51180	0.37500	0.25000	0.23542
0.75	Action projection	5	0.51131	0.31250	0.21875	0.20020
0.75	Action projection	6	0.53841	0.46875	0.28125	0.30000
0.75	Action projection	7	0.52215	0.46875	0.31250	0.29071
0.75	Action projection	8	0.54017	0.53125	0.21875	0.36352
0.75	Action projection	9	0.52716	0.43750	0.15625	0.32044
0.75	Calibrated tree	0	0.77123	0.68750	0.15625	0.58101
0.75	Calibrated tree	1	0.75264	0.59375	0.18750	0.49701
0.75	Calibrated tree	2	0.72495	0.65625	0.12500	0.55452
0.75	Calibrated tree	3	0.75342	0.56250	0.09375	0.51178
0.75	Calibrated tree	4	0.71875	0.56250	0.18750	0.46269
0.75	Calibrated tree	5	0.73596	0.56250	0.18750	0.47036
0.75	Calibrated tree	6	0.74079	0.81250	0.12500	0.66884
0.75	Calibrated tree	7	0.73656	0.53125	0.06250	0.48941
0.75	Calibrated tree	8	0.72209	0.62500	0.18750	0.50542
0.75	Calibrated tree	9	0.74534	0.50000	0.03125	0.48508
0.75	Causal flow v5	0	0.76229	0.65625	0.15625	0.54721

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
1080	0.75 Causal flow v5	1	0.74222	0.40625	0.03125	0.41232
1081	0.75 Causal flow v5	2	0.71861	0.59375	0.12500	0.50048
1082	0.75 Causal flow v5	3	0.73177	0.43750	0.03125	0.43481
1083	0.75 Causal flow v5	4	0.72471	0.62500	0.03125	0.55916
1084	0.75 Causal flow v5	5	0.72510	0.56250	0.03125	0.51872
1085	0.75 Causal flow v5	6	0.72175	0.43750	0.09375	0.40586
1086	0.75 Causal flow v5	7	0.70493	0.56250	0.12500	0.47268
1087	0.75 Causal flow v5	8	0.71894	0.62500	0.03125	0.55810
1088	0.75 Causal flow v5	9	0.73616	0.59375	0.15625	0.49726
1089	0.75 Contrastive action	0	0.73626	0.78125	0.15625	0.64657
1090	0.75 Contrastive action	1	0.72159	0.59375	0.15625	0.50908
1091	0.75 Contrastive action	2	0.67798	0.59375	0.21875	0.46896
1092	0.75 Contrastive action	3	0.73454	0.68750	0.18750	0.56873
1093	0.75 Contrastive action	4	0.69205	0.53125	0.15625	0.45255
1094	0.75 Contrastive action	5	0.68529	0.62500	0.06250	0.54880
1095	0.75 Contrastive action	6	0.70283	0.53125	0.06250	0.49107
1096	0.75 Contrastive action	7	0.70875	0.62500	0.15625	0.52436
1097	0.75 Contrastive action	8	0.68668	0.62500	0.21875	0.49073
1098	0.75 Contrastive action	9	0.71626	0.65625	0.00000	0.60528
1099	0.75 Learned correlation	0	0.75341	0.50000	0.12500	0.45491
1100	0.75 Learned correlation	1	0.73325	0.65625	0.15625	0.54240
1101	0.75 Learned correlation	2	0.69232	0.50000	0.21875	0.39602
1102	0.75 Learned correlation	3	0.73816	0.46875	0.25000	0.38582
1103	0.75 Learned correlation	4	0.69375	0.53125	0.15625	0.43965
1104	0.75 Learned correlation	5	0.70220	0.62500	0.21875	0.48695
1105	0.75 Learned correlation	6	0.72328	0.68750	0.12500	0.57331
1106	0.75 Learned correlation	7	0.72304	0.71875	0.06250	0.61595
1107	0.75 Learned correlation	8	0.69972	0.59375	0.12500	0.49613
1108	0.75 Learned correlation	9	0.71716	0.65625	0.18750	0.52759
1109	0.75 Oracle	0	1.00000	1.00000	0.06250	0.95750
1110	0.75 Oracle	1	1.00000	1.00000	0.03125	0.96875
1111	0.75 Oracle	2	1.00000	0.96875	0.03125	0.94687
1112	0.75 Oracle	3	1.00000	1.00000	0.00000	0.98000
1113	0.75 Oracle	4	1.00000	0.93750	0.00000	0.93625
1114	0.75 Oracle	5	1.00000	0.96875	0.03125	0.94688
1115	0.75 Oracle	6	1.00000	0.96875	0.00000	0.95813
1116	0.75 Oracle	7	1.00000	0.96875	0.00000	0.95813
1117	0.75 Oracle	8	1.00000	0.96875	0.00000	0.95813
1118	0.75 Oracle	9	1.00000	0.96875	0.03125	0.94687
1119	0.75 World model	0	0.75681	0.65625	0.09375	0.57040
1120	0.75 World model	1	0.74990	0.81250	0.06250	0.69240
1121	0.75 World model	2	0.72113	0.53125	0.18750	0.43606
1122	0.75 World model	3	0.74130	0.53125	0.12500	0.46928
1123	0.75 World model	4	0.73396	0.50000	0.06250	0.46520
1124	0.75 World model	5	0.72266	0.62500	0.18750	0.50750
1125	0.75 World model	6	0.72230	0.65625	0.25000	0.50266
1126	0.75 World model	7	0.72396	0.65625	0.21875	0.51418
1127	0.75 World model	8	0.73078	0.56250	0.25000	0.43958
1128	0.75 World model	9	0.74690	0.59375	0.06250	0.53689
1129	1.00 Action projection	0	0.45622	0.31250	0.28125	0.15217
1130	1.00 Action projection	1	0.45091	0.15625	0.15625	0.08772
1131	1.00 Action projection	2	0.45388	0.40625	0.12500	0.27820
1132	1.00 Action projection	3	0.46695	0.31250	0.25000	0.17980
1133	1.00 Action projection	4	0.45998	0.25000	0.25000	0.13834
1134	1.00 Action projection	5	0.45724	0.25000	0.25000	0.11787
1135	1.00 Action projection	6	0.48010	0.40625	0.31250	0.22372
1136	1.00 Action projection	7	0.47569	0.46875	0.25000	0.28357
1137	1.00 Action projection	8	0.45992	0.46875	0.31250	0.26517
1138	1.00 Action projection	9	0.47306	0.31250	0.18750	0.19034
1139	1.00 Calibrated tree	0	0.68712	0.53125	0.21875	0.41185
1140	1.00 Calibrated tree	1	0.67282	0.53125	0.28125	0.37835
1141	1.00 Calibrated tree	2	0.67488	0.62500	0.18750	0.48674
1142	1.00 Calibrated tree	3	0.70239	0.56250	0.12500	0.47725
1143	1.00 Calibrated tree	4	0.69591	0.53125	0.15625	0.43899
1144	1.00 Calibrated tree	5	0.68181	0.43750	0.25000	0.32952
1145	1.00 Calibrated tree	6	0.69796	0.78125	0.12500	0.62275
1146	1.00 Calibrated tree	7	0.69797	0.46875	0.09375	0.41587
1147	1.00 Calibrated tree	8	0.69557	0.68750	0.25000	0.51425
1148	1.00 Calibrated tree	9	0.70174	0.53125	0.03125	0.48283
1149	1.00 Causal flow v5	0	0.67208	0.65625	0.15625	0.51338
1150	1.00 Causal flow v5	1	0.65154	0.25000	0.03125	0.25028
1151	1.00 Causal flow v5	2	0.64287	0.50000	0.15625	0.38850
1152	1.00 Causal flow v5	3	0.68180	0.37500	0.03125	0.36750
1153	1.00 Causal flow v5	4	0.66034	0.53125	0.03125	0.46132
1154	1.00 Causal flow v5	5	0.66918	0.43750	0.06250	0.38848
1155	1.00 Causal flow v5	6	0.67233	0.46875	0.12500	0.39122
1156	1.00 Causal flow v5	7	0.68312	0.46875	0.12500	0.39596

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
1.00	Causal flow v5	8	0.65260	0.56250	0.09375	0.45861
1.00	Causal flow v5	9	0.67564	0.53125	0.15625	0.42306
1.00	Contrastive action	0	0.61476	0.46875	0.15625	0.37219
1.00	Contrastive action	1	0.61906	0.56250	0.18750	0.42466
1.00	Contrastive action	2	0.62455	0.59375	0.18750	0.45886
1.00	Contrastive action	3	0.65963	0.59375	0.15625	0.48515
1.00	Contrastive action	4	0.64680	0.53125	0.18750	0.42445
1.00	Contrastive action	5	0.62553	0.50000	0.06250	0.43149
1.00	Contrastive action	6	0.65567	0.40625	0.12500	0.35570
1.00	Contrastive action	7	0.63026	0.50000	0.21875	0.37693
1.00	Contrastive action	8	0.63362	0.56250	0.28125	0.40451
1.00	Contrastive action	9	0.65011	0.56250	0.00000	0.50832
1.00	Learned correlation	0	0.65773	0.37500	0.15625	0.31079
1.00	Learned correlation	1	0.64254	0.43750	0.15625	0.34481
1.00	Learned correlation	2	0.64398	0.37500	0.25000	0.27834
1.00	Learned correlation	3	0.68156	0.43750	0.25000	0.33607
1.00	Learned correlation	4	0.67356	0.46875	0.15625	0.38664
1.00	Learned correlation	5	0.64820	0.59375	0.21875	0.43657
1.00	Learned correlation	6	0.68038	0.53125	0.09375	0.45248
1.00	Learned correlation	7	0.66380	0.59375	0.12500	0.47781
1.00	Learned correlation	8	0.66817	0.46875	0.21875	0.36235
1.00	Learned correlation	9	0.67330	0.56250	0.31250	0.39070
1.00	Oracle	0	1.00000	1.00000	0.06250	0.95750
1.00	Oracle	1	1.00000	1.00000	0.06250	0.95750
1.00	Oracle	2	1.00000	0.96875	0.03125	0.94687
1.00	Oracle	3	1.00000	1.00000	0.00000	0.98000
1.00	Oracle	4	1.00000	0.93750	0.00000	0.93625
1.00	Oracle	5	1.00000	0.96875	0.03125	0.94688
1.00	Oracle	6	1.00000	0.96875	0.00000	0.95813
1.00	Oracle	7	1.00000	0.96875	0.00000	0.95813
1.00	Oracle	8	1.00000	0.96875	0.00000	0.95813
1.00	Oracle	9	1.00000	0.96875	0.06250	0.93562
1.00	World model	0	0.66948	0.50000	0.12500	0.41328
1.00	World model	1	0.65713	0.71875	0.09375	0.56563
1.00	World model	2	0.65411	0.43750	0.18750	0.33857
1.00	World model	3	0.68080	0.46875	0.12500	0.40047
1.00	World model	4	0.66531	0.40625	0.03125	0.37582
1.00	World model	5	0.67578	0.59375	0.18750	0.46100
1.00	World model	6	0.68097	0.59375	0.28125	0.42814
1.00	World model	7	0.68187	0.50000	0.21875	0.38669
1.00	World model	8	0.66446	0.46875	0.25000	0.34000
1.00	World model	9	0.67982	0.50000	0.06250	0.43934
1.25	Action projection	0	0.38622	0.28125	0.25000	0.10915
1.25	Action projection	1	0.41500	0.21875	0.18750	0.10161
1.25	Action projection	2	0.39798	0.28125	0.15625	0.14506
1.25	Action projection	3	0.40041	0.31250	0.25000	0.13673
1.25	Action projection	4	0.40232	0.25000	0.25000	0.09336
1.25	Action projection	5	0.40970	0.21875	0.31250	0.06008
1.25	Action projection	6	0.39403	0.15625	0.34375	0.01094
1.25	Action projection	7	0.38659	0.28125	0.40625	0.04510
1.25	Action projection	8	0.40715	0.37500	0.28125	0.17607
1.25	Action projection	9	0.40496	0.34375	0.28125	0.14837
1.25	Calibrated tree	0	0.63167	0.37500	0.21875	0.26577
1.25	Calibrated tree	1	0.62635	0.46875	0.25000	0.32003
1.25	Calibrated tree	2	0.61330	0.56250	0.21875	0.38769
1.25	Calibrated tree	3	0.62251	0.43750	0.12500	0.34128
1.25	Calibrated tree	4	0.62250	0.43750	0.12500	0.34592
1.25	Calibrated tree	5	0.60706	0.31250	0.28125	0.19383
1.25	Calibrated tree	6	0.64535	0.71875	0.12500	0.55369
1.25	Calibrated tree	7	0.61435	0.34375	0.12500	0.27369
1.25	Calibrated tree	8	0.64021	0.50000	0.28125	0.33304
1.25	Calibrated tree	9	0.64871	0.34375	0.06250	0.30584
1.25	Causal flow v5	0	0.62034	0.56250	0.18750	0.39702
1.25	Causal flow v5	1	0.61447	0.25000	0.03125	0.23494
1.25	Causal flow v5	2	0.61930	0.43750	0.15625	0.32191
1.25	Causal flow v5	3	0.60085	0.31250	0.09375	0.24373
1.25	Causal flow v5	4	0.59377	0.40625	0.09375	0.31903
1.25	Causal flow v5	5	0.58062	0.37500	0.09375	0.28699
1.25	Causal flow v5	6	0.61066	0.34375	0.15625	0.25584
1.25	Causal flow v5	7	0.59781	0.46875	0.12500	0.35348
1.25	Causal flow v5	8	0.61363	0.53125	0.06250	0.42431
1.25	Causal flow v5	9	0.63512	0.40625	0.15625	0.29860
1.25	Contrastive action	0	0.54131	0.43750	0.31250	0.25609
1.25	Contrastive action	1	0.54400	0.43750	0.18750	0.29763
1.25	Contrastive action	2	0.51502	0.31250	0.21875	0.18453
1.25	Contrastive action	3	0.53432	0.37500	0.21875	0.23899
1.25	Contrastive action	4	0.53953	0.34375	0.25000	0.21111

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
1.25	Contrastive action	5	0.53751	0.40625	0.06250	0.31859
1.25	Contrastive action	6	0.56619	0.34375	0.12500	0.27757
1.25	Contrastive action	7	0.51625	0.40625	0.28125	0.23265
1.25	Contrastive action	8	0.56240	0.43750	0.31250	0.26517
1.25	Contrastive action	9	0.56307	0.43750	0.06250	0.35595
1.25	Learned correlation	0	0.59589	0.31250	0.18750	0.22056
1.25	Learned correlation	1	0.59211	0.37500	0.15625	0.27309
1.25	Learned correlation	2	0.56908	0.28125	0.34375	0.12712
1.25	Learned correlation	3	0.58672	0.28125	0.31250	0.14875
1.25	Learned correlation	4	0.58433	0.28125	0.28125	0.16319
1.25	Learned correlation	5	0.57667	0.43750	0.21875	0.28959
1.25	Learned correlation	6	0.61232	0.43750	0.12500	0.34814
1.25	Learned correlation	7	0.57001	0.53125	0.18750	0.36167
1.25	Learned correlation	8	0.59889	0.46875	0.18750	0.33010
1.25	Learned correlation	9	0.61430	0.46875	0.31250	0.29050
1.25	Oracle	0	1.00000	1.00000	0.06250	0.95750
1.25	Oracle	1	1.00000	1.00000	0.06250	0.95750
1.25	Oracle	2	1.00000	0.96875	0.03125	0.94687
1.25	Oracle	3	1.00000	1.00000	0.00000	0.98000
1.25	Oracle	4	1.00000	0.93750	0.00000	0.93625
1.25	Oracle	5	1.00000	0.96875	0.03125	0.94688
1.25	Oracle	6	1.00000	0.96875	0.00000	0.95813
1.25	Oracle	7	1.00000	0.96875	0.00000	0.95813
1.25	Oracle	8	1.00000	0.96875	0.00000	0.95813
1.25	Oracle	9	1.00000	0.96875	0.06250	0.93562
1.25	World model	0	0.61410	0.53125	0.12500	0.40203
1.25	World model	1	0.59202	0.56250	0.12500	0.41725
1.25	World model	2	0.60403	0.25000	0.18750	0.17977
1.25	World model	3	0.59227	0.25000	0.21875	0.15424
1.25	World model	4	0.59563	0.40625	0.06250	0.33239
1.25	World model	5	0.59140	0.43750	0.25000	0.27971
1.25	World model	6	0.62381	0.50000	0.31250	0.31687
1.25	World model	7	0.60108	0.46875	0.21875	0.32171
1.25	World model	8	0.61147	0.53125	0.28125	0.34603
1.25	World model	9	0.63855	0.50000	0.06250	0.40346
1.50	Action projection	0	0.38167	0.21875	0.25000	0.06888
1.50	Action projection	1	0.37180	0.09375	0.18750	-0.00942
1.50	Action projection	2	0.37434	0.21875	0.21875	0.06627
1.50	Action projection	3	0.36216	0.25000	0.28125	0.05348
1.50	Action projection	4	0.37014	0.21875	0.21875	0.07770
1.50	Action projection	5	0.35744	0.18750	0.31250	-0.00071
1.50	Action projection	6	0.37985	0.21875	0.40625	-0.00123
1.50	Action projection	7	0.36506	0.28125	0.37500	0.05150
1.50	Action projection	8	0.37019	0.31250	0.34375	0.08895
1.50	Action projection	9	0.37764	0.21875	0.28125	0.04909
1.50	Calibrated tree	0	0.61442	0.40625	0.21875	0.28338
1.50	Calibrated tree	1	0.55554	0.28125	0.31250	0.12186
1.50	Calibrated tree	2	0.61506	0.59375	0.21875	0.40597
1.50	Calibrated tree	3	0.59207	0.34375	0.21875	0.21564
1.50	Calibrated tree	4	0.60295	0.31250	0.21875	0.20596
1.50	Calibrated tree	5	0.58171	0.28125	0.28125	0.14780
1.50	Calibrated tree	6	0.59025	0.59375	0.18750	0.39661
1.50	Calibrated tree	7	0.59142	0.37500	0.21875	0.23746
1.50	Calibrated tree	8	0.58133	0.37500	0.25000	0.22868
1.50	Calibrated tree	9	0.60396	0.28125	0.09375	0.22440
1.50	Causal flow v5	0	0.58835	0.46875	0.15625	0.32145
1.50	Causal flow v5	1	0.53459	0.15625	0.03125	0.11861
1.50	Causal flow v5	2	0.58596	0.40625	0.15625	0.27270
1.50	Causal flow v5	3	0.57479	0.37500	0.09375	0.27430
1.50	Causal flow v5	4	0.55329	0.40625	0.09375	0.29003
1.50	Causal flow v5	5	0.58483	0.34375	0.09375	0.25645
1.50	Causal flow v5	6	0.55622	0.18750	0.25000	0.07560
1.50	Causal flow v5	7	0.57200	0.40625	0.15625	0.27003
1.50	Causal flow v5	8	0.56104	0.34375	0.09375	0.24672
1.50	Causal flow v5	9	0.58238	0.40625	0.15625	0.27429
1.50	Contrastive action	0	0.51889	0.50000	0.21875	0.32297
1.50	Contrastive action	1	0.46712	0.25000	0.31250	0.07444
1.50	Contrastive action	2	0.50658	0.25000	0.21875	0.13528
1.50	Contrastive action	3	0.49783	0.28125	0.21875	0.14322
1.50	Contrastive action	4	0.49945	0.37500	0.25000	0.21081
1.50	Contrastive action	5	0.48954	0.31250	0.06250	0.22036
1.50	Contrastive action	6	0.48787	0.15625	0.15625	0.07706
1.50	Contrastive action	7	0.49777	0.37500	0.25000	0.20221
1.50	Contrastive action	8	0.49441	0.34375	0.34375	0.14730
1.50	Contrastive action	9	0.49379	0.31250	0.03125	0.23960
1.50	Learned correlation	0	0.56711	0.18750	0.18750	0.12260
1.50	Learned correlation	1	0.52582	0.34375	0.18750	0.19871

Table 10: Seed-level combined stress metrics. (continued)

Level	Method	Seed	F1	Success	Unsafe	Utility
1.50	Learned correlation	2	0.56945	0.28125	0.40625	0.10142
1.50	Learned correlation	3	0.55099	0.18750	0.34375	0.04221
1.50	Learned correlation	4	0.56337	0.34375	0.21875	0.21226
1.50	Learned correlation	5	0.53698	0.40625	0.25000	0.22473
1.50	Learned correlation	6	0.54758	0.31250	0.25000	0.16211
1.50	Learned correlation	7	0.55547	0.43750	0.18750	0.27764
1.50	Learned correlation	8	0.54857	0.37500	0.28125	0.20274
1.50	Learned correlation	9	0.55704	0.37500	0.37500	0.17259
1.50	Oracle	0	1.00000	1.00000	0.09375	0.94625
1.50	Oracle	1	1.00000	1.00000	0.06250	0.95750
1.50	Oracle	2	1.00000	0.90625	0.03125	0.90312
1.50	Oracle	3	1.00000	0.96875	0.00000	0.95813
1.50	Oracle	4	1.00000	0.93750	0.00000	0.93625
1.50	Oracle	5	1.00000	0.93750	0.03125	0.92500
1.50	Oracle	6	1.00000	0.96875	0.00000	0.95813
1.50	Oracle	7	1.00000	0.96875	0.00000	0.95813
1.50	Oracle	8	1.00000	0.87500	0.00000	0.89250
1.50	Oracle	9	1.00000	0.96875	0.06250	0.93562
1.50	World model	0	0.59315	0.40625	0.12500	0.29359
1.50	World model	1	0.52211	0.43750	0.12500	0.28504
1.50	World model	2	0.58051	0.31250	0.18750	0.19646
1.50	World model	3	0.57552	0.21875	0.21875	0.12561
1.50	World model	4	0.54647	0.34375	0.06250	0.25766
1.50	World model	5	0.57641	0.40625	0.21875	0.25766
1.50	World model	6	0.54226	0.37500	0.34375	0.17147
1.50	World model	7	0.56910	0.43750	0.25000	0.26390
1.50	World model	8	0.55176	0.37500	0.31250	0.19072
1.50	World model	9	0.57661	0.43750	0.06250	0.33347

E ABLATION SEED METRICS

Table 11: Seed-level ablation metrics.

Split	Ablation	Seed	F1	Success	Unsafe	Utility
combined_hard_shift	correlation_only_causal_head	0	0.72998	0.56250	0.08750	0.49180
combined_hard_shift	correlation_only_causal_head	1	0.73043	0.67500	0.13750	0.55239
combined_hard_shift	correlation_only_causal_head	2	0.73786	0.52500	0.11250	0.45971
combined_hard_shift	correlation_only_causal_head	3	0.76249	0.50000	0.15000	0.43867
combined_hard_shift	correlation_only_causal_head	4	0.74699	0.50000	0.20000	0.41317
combined_hard_shift	correlation_only_causal_head	5	0.71577	0.46250	0.20000	0.37480
combined_hard_shift	correlation_only_causal_head	6	0.72875	0.51250	0.10000	0.44922
combined_hard_shift	correlation_only_causal_head	7	0.75935	0.61250	0.11250	0.52985
combined_hard_shift	correlation_only_causal_head	8	0.73543	0.61250	0.17500	0.49871
combined_hard_shift	correlation_only_causal_head	9	0.75244	0.61250	0.07500	0.54214
combined_hard_shift	full_causal_scene_flow_model_v5	0	0.75509	0.66250	0.05000	0.59149
combined_hard_shift	full_causal_scene_flow_model_v5	1	0.74523	0.61250	0.12500	0.52985
combined_hard_shift	full_causal_scene_flow_model_v5	2	0.74399	0.55000	0.07500	0.50204
combined_hard_shift	full_causal_scene_flow_model_v5	3	0.76736	0.62500	0.07500	0.56068
combined_hard_shift	full_causal_scene_flow_model_v5	4	0.76920	0.62500	0.08750	0.55846
combined_hard_shift	full_causal_scene_flow_model_v5	5	0.75041	0.61250	0.06250	0.55263
combined_hard_shift	full_causal_scene_flow_model_v5	6	0.75668	0.70000	0.08750	0.60503
combined_hard_shift	full_causal_scene_flow_model_v5	7	0.77278	0.63750	0.06250	0.57926
combined_hard_shift	full_causal_scene_flow_model_v5	8	0.75005	0.57500	0.07500	0.52302
combined_hard_shift	full_causal_scene_flow_model_v5	9	0.75410	0.61250	0.12500	0.53307
combined_hard_shift	mask_only_no_effect_predictor	0	0.75509	0.43750	0.12500	0.37939
combined_hard_shift	mask_only_no_effect_predictor	1	0.74523	0.40000	0.12500	0.35350
combined_hard_shift	mask_only_no_effect_predictor	2	0.74399	0.45000	0.17500	0.36844
combined_hard_shift	mask_only_no_effect_predictor	3	0.76736	0.48750	0.11250	0.42333
combined_hard_shift	mask_only_no_effect_predictor	4	0.76920	0.38750	0.07500	0.36911
combined_hard_shift	mask_only_no_effect_predictor	5	0.75041	0.51250	0.17500	0.41453
combined_hard_shift	mask_only_no_effect_predictor	6	0.75668	0.45000	0.21250	0.35743
combined_hard_shift	mask_only_no_effect_predictor	7	0.77278	0.48750	0.12500	0.42416
combined_hard_shift	mask_only_no_effect_predictor	8	0.75005	0.50000	0.16250	0.41142
combined_hard_shift	mask_only_no_effect_predictor	9	0.75410	0.38750	0.12500	0.34797
combined_hard_shift	minus_action_conditioning	0	0.74267	0.53750	0.06250	0.47864
combined_hard_shift	minus_action_conditioning	1	0.74898	0.53750	0.13750	0.45584
combined_hard_shift	minus_action_conditioning	2	0.74893	0.63750	0.07500	0.54790
combined_hard_shift	minus_action_conditioning	3	0.76005	0.55000	0.15000	0.46156
combined_hard_shift	minus_action_conditioning	4	0.75745	0.53750	0.16250	0.44617

Table 11: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	F1	Success	Unsafe	Utility
combined_hard_shift	minus_action_conditioning	5	0.73865	0.50000	0.11250	0.43331
combined_hard_shift	minus_action_conditioning	6	0.75373	0.56250	0.16250	0.46287
combined_hard_shift	minus_action_conditioning	7	0.76552	0.62500	0.15000	0.51866
combined_hard_shift	minus_action_conditioning	8	0.74864	0.61250	0.15000	0.50394
combined_hard_shift	minus_action_conditioning	9	0.75389	0.53750	0.08750	0.47641
combined_hard_shift	minus_articulation_graph	0	0.71796	0.55000	0.10000	0.49471
combined_hard_shift	minus_articulation_graph	1	0.69359	0.68750	0.13750	0.57307
combined_hard_shift	minus_articulation_graph	2	0.70548	0.53750	0.10000	0.48224
combined_hard_shift	minus_articulation_graph	3	0.73251	0.60000	0.16250	0.51138
combined_hard_shift	minus_articulation_graph	4	0.70902	0.56250	0.15000	0.48440
combined_hard_shift	minus_articulation_graph	5	0.70235	0.52500	0.17500	0.44755
combined_hard_shift	minus_articulation_graph	6	0.70827	0.53750	0.11250	0.47984
combined_hard_shift	minus_articulation_graph	7	0.74010	0.53750	0.12500	0.48684
combined_hard_shift	minus_articulation_graph	8	0.70828	0.62500	0.16250	0.52365
combined_hard_shift	minus_articulation_graph	9	0.72289	0.60000	0.15000	0.51375
combined_hard_shift	minus_counterfactual_negatives	0	0.75583	0.55000	0.15000	0.47186
combined_hard_shift	minus_counterfactual_negatives	1	0.74550	0.51250	0.15000	0.44606
combined_hard_shift	minus_counterfactual_negatives	2	0.74718	0.70000	0.13750	0.58029
combined_hard_shift	minus_counterfactual_negatives	3	0.76698	0.65000	0.07500	0.57285
combined_hard_shift	minus_counterfactual_negatives	4	0.76606	0.53750	0.15000	0.46833
combined_hard_shift	minus_counterfactual_negatives	5	0.74803	0.60000	0.11250	0.51965
combined_hard_shift	minus_counterfactual_negatives	6	0.75246	0.57500	0.07500	0.51552
combined_hard_shift	minus_counterfactual_negatives	7	0.77479	0.58750	0.12500	0.51706
combined_hard_shift	minus_counterfactual_negatives	8	0.74773	0.60000	0.15000	0.50823
combined_hard_shift	minus_counterfactual_negatives	9	0.75661	0.55000	0.07500	0.50365
combined_hard_shift	minus_occlusion_reasoning	0	0.74551	0.55000	0.13750	0.47962
combined_hard_shift	minus_occlusion_reasoning	1	0.73902	0.63750	0.12500	0.54616
combined_hard_shift	minus_occlusion_reasoning	2	0.73642	0.60000	0.16250	0.50435
combined_hard_shift	minus_occlusion_reasoning	3	0.75967	0.45000	0.08750	0.43210
combined_hard_shift	minus_occlusion_reasoning	4	0.75752	0.56250	0.08750	0.51130
combined_hard_shift	minus_occlusion_reasoning	5	0.74216	0.57500	0.16250	0.48899
combined_hard_shift	minus_occlusion_reasoning	6	0.74857	0.58750	0.16250	0.49791
combined_hard_shift	minus_occlusion_reasoning	7	0.76428	0.63750	0.10000	0.56395
combined_hard_shift	minus_occlusion_reasoning	8	0.74449	0.50000	0.11250	0.45634
combined_hard_shift	minus_occlusion_reasoning	9	0.75149	0.62500	0.10000	0.55036
combined_hard_shift	minus_passive_flow_factor	0	0.70788	0.48750	0.31250	0.33363
combined_hard_shift	minus_passive_flow_factor	1	0.71502	0.43750	0.16250	0.35722
combined_hard_shift	minus_passive_flow_factor	2	0.71471	0.50000	0.12500	0.41453
combined_hard_shift	minus_passive_flow_factor	3	0.73475	0.55000	0.18750	0.43545
combined_hard_shift	minus_passive_flow_factor	4	0.71629	0.48750	0.13750	0.40014
combined_hard_shift	minus_passive_flow_factor	5	0.69996	0.47500	0.13750	0.38370
combined_hard_shift	minus_passive_flow_factor	6	0.70021	0.51250	0.22500	0.37682
combined_hard_shift	minus_passive_flow_factor	7	0.72455	0.46250	0.10000	0.40069
combined_hard_shift	minus_passive_flow_factor	8	0.70673	0.50000	0.13750	0.40743
combined_hard_shift	minus_passive_flow_factor	9	0.72555	0.56250	0.22500	0.43004
combined_hard_shift	minus_uncertainty_calibration	0	0.74269	0.56250	0.18750	0.46465
combined_hard_shift	minus_uncertainty_calibration	1	0.72896	0.53750	0.21250	0.43812
combined_hard_shift	minus_uncertainty_calibration	2	0.73183	0.67500	0.15000	0.55593
combined_hard_shift	minus_uncertainty_calibration	3	0.75366	0.56250	0.17500	0.47374
combined_hard_shift	minus_uncertainty_calibration	4	0.75170	0.61250	0.13750	0.52290
combined_hard_shift	minus_uncertainty_calibration	5	0.73274	0.65000	0.21250	0.51651
combined_hard_shift	minus_uncertainty_calibration	6	0.74025	0.60000	0.20000	0.48600
combined_hard_shift	minus_uncertainty_calibration	7	0.76294	0.73750	0.11250	0.62456
combined_hard_shift	minus_uncertainty_calibration	8	0.73236	0.61250	0.15000	0.51398
combined_hard_shift	minus_uncertainty_calibration	9	0.74273	0.60000	0.21250	0.48744
combined_hard_shift	old_v4_causal_score	0	0.76323	0.55000	0.15000	0.48007
combined_hard_shift	old_v4_causal_score	1	0.75738	0.61250	0.10000	0.54190
combined_hard_shift	old_v4_causal_score	2	0.75802	0.67500	0.10000	0.58490
combined_hard_shift	old_v4_causal_score	3	0.78095	0.70000	0.17500	0.58144
combined_hard_shift	old_v4_causal_score	4	0.78189	0.65000	0.18750	0.54299
combined_hard_shift	old_v4_causal_score	5	0.76456	0.66250	0.10000	0.57849
combined_hard_shift	old_v4_causal_score	6	0.76687	0.58750	0.12500	0.51627
combined_hard_shift	old_v4_causal_score	7	0.78456	0.60000	0.07500	0.55264
combined_hard_shift	old_v4_causal_score	8	0.76467	0.63750	0.12500	0.55342
combined_hard_shift	old_v4_causal_score	9	0.77053	0.60000	0.16250	0.51563
distractor_contact_shift	correlation_only_causal_head	0	0.77319	0.72500	0.16250	0.59995
distractor_contact_shift	correlation_only_causal_head	1	0.77035	0.58750	0.16250	0.50350
distractor_contact_shift	correlation_only_causal_head	2	0.75838	0.66250	0.16250	0.54860
distractor_contact_shift	correlation_only_causal_head	3	0.76529	0.61250	0.20000	0.50412
distractor_contact_shift	correlation_only_causal_head	4	0.77230	0.63750	0.12500	0.55003
distractor_contact_shift	correlation_only_causal_head	5	0.77543	0.65000	0.18750	0.54006
distractor_contact_shift	correlation_only_causal_head	6	0.76794	0.56250	0.11250	0.50117
distractor_contact_shift	correlation_only_causal_head	7	0.77571	0.68750	0.15000	0.57915
distractor_contact_shift	correlation_only_causal_head	8	0.77521	0.61250	0.16250	0.52130
distractor_contact_shift	correlation_only_causal_head	9	0.75412	0.65000	0.21250	0.52146
distractor_contact_shift	full_causal_scene_flow_model_v5	0	0.78053	0.67500	0.05000	0.61443
distractor_contact_shift	full_causal_scene_flow_model_v5	1	0.76893	0.53750	0.13750	0.48159

Table 11: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	F1	Success	Unsafe	Utility
distractor_contact_shift	full_causal_scene_flow_model.v5	2	0.77418	0.68750	0.06250	0.61536
distractor_contact_shift	full_causal_scene_flow_model.v5	3	0.76701	0.66250	0.13750	0.56878
distractor_contact_shift	full_causal_scene_flow_model.v5	4	0.77765	0.52500	0.12500	0.47997
distractor_contact_shift	full_causal_scene_flow_model.v5	5	0.78064	0.71250	0.17500	0.59642
distractor_contact_shift	full_causal_scene_flow_model.v5	6	0.77383	0.67500	0.08750	0.59765
distractor_contact_shift	full_causal_scene_flow_model.v5	7	0.77619	0.78750	0.11250	0.66944
distractor_contact_shift	full_causal_scene_flow_model.v5	8	0.78004	0.66250	0.21250	0.54686
distractor_contact_shift	full_causal_scene_flow_model.v5	9	0.76278	0.56250	0.06250	0.52554
distractor_contact_shift	mask_only_no_effect_predictor	0	0.78053	0.48750	0.16250	0.41508
distractor_contact_shift	mask_only_no_effect_predictor	1	0.76893	0.53750	0.05000	0.48549
distractor_contact_shift	mask_only_no_effect_predictor	2	0.77418	0.37500	0.12500	0.34651
distractor_contact_shift	mask_only_no_effect_predictor	3	0.76701	0.48750	0.20000	0.39618
distractor_contact_shift	mask_only_no_effect_predictor	4	0.77765	0.46250	0.12500	0.40862
distractor_contact_shift	mask_only_no_effect_predictor	5	0.78064	0.50000	0.15000	0.42907
distractor_contact_shift	mask_only_no_effect_predictor	6	0.77383	0.38750	0.12500	0.35530
distractor_contact_shift	mask_only_no_effect_predictor	7	0.77619	0.52500	0.12500	0.45359
distractor_contact_shift	mask_only_no_effect_predictor	8	0.78004	0.41250	0.16250	0.36226
distractor_contact_shift	mask_only_no_effect_predictor	9	0.76278	0.52500	0.12500	0.44919
distractor_contact_shift	minus_action_conditioning	0	0.73354	0.51250	0.13750	0.43534
distractor_contact_shift	minus_action_conditioning	1	0.73156	0.61250	0.06250	0.53052
distractor_contact_shift	minus_action_conditioning	2	0.73113	0.60000	0.20000	0.47012
distractor_contact_shift	minus_action_conditioning	3	0.73249	0.53750	0.13750	0.45160
distractor_contact_shift	minus_action_conditioning	4	0.75135	0.67500	0.13750	0.55532
distractor_contact_shift	minus_action_conditioning	5	0.73811	0.47500	0.21250	0.38373
distractor_contact_shift	minus_action_conditioning	6	0.74358	0.50000	0.10000	0.44160
distractor_contact_shift	minus_action_conditioning	7	0.73705	0.65000	0.20000	0.51052
distractor_contact_shift	minus_action_conditioning	8	0.73500	0.42500	0.12500	0.37750
distractor_contact_shift	minus_action_conditioning	9	0.73790	0.62500	0.18750	0.49925
distractor_contact_shift	minus_articulation_graph	0	0.76852	0.62500	0.07500	0.57476
distractor_contact_shift	minus_articulation_graph	1	0.76390	0.71250	0.11250	0.61944
distractor_contact_shift	minus_articulation_graph	2	0.76226	0.72500	0.11250	0.62890
distractor_contact_shift	minus_articulation_graph	3	0.77707	0.57500	0.07500	0.54004
distractor_contact_shift	minus_articulation_graph	4	0.77361	0.65000	0.12500	0.57531
distractor_contact_shift	minus_articulation_graph	5	0.76793	0.68750	0.11250	0.60382
distractor_contact_shift	minus_articulation_graph	6	0.75389	0.43750	0.12500	0.41833
distractor_contact_shift	minus_articulation_graph	7	0.77049	0.72500	0.06250	0.64811
distractor_contact_shift	minus_articulation_graph	8	0.76550	0.63750	0.10000	0.57106
distractor_contact_shift	minus_articulation_graph	9	0.75149	0.65000	0.10000	0.57926
distractor_contact_shift	minus_counterfactual_negatives	0	0.77597	0.66250	0.08750	0.58671
distractor_contact_shift	minus_counterfactual_negatives	1	0.76275	0.68750	0.17500	0.56658
distractor_contact_shift	minus_counterfactual_negatives	2	0.76107	0.60000	0.08750	0.53516
distractor_contact_shift	minus_counterfactual_negatives	3	0.76020	0.63750	0.12500	0.54856
distractor_contact_shift	minus_counterfactual_negatives	4	0.77277	0.68750	0.10000	0.59651
distractor_contact_shift	minus_counterfactual_negatives	5	0.77002	0.65000	0.17500	0.54392
distractor_contact_shift	minus_counterfactual_negatives	6	0.76476	0.55000	0.07500	0.50696
distractor_contact_shift	minus_counterfactual_negatives	7	0.76390	0.65000	0.15000	0.55012
distractor_contact_shift	minus_counterfactual_negatives	8	0.77232	0.53750	0.11250	0.48805
distractor_contact_shift	minus_counterfactual_negatives	9	0.75446	0.71250	0.16250	0.58646
distractor_contact_shift	minus_occlusion_reasoning	0	0.77567	0.66250	0.17500	0.56022
distractor_contact_shift	minus_occlusion_reasoning	1	0.76441	0.63750	0.12500	0.55521
distractor_contact_shift	minus_occlusion_reasoning	2	0.76666	0.58750	0.11250	0.52625
distractor_contact_shift	minus_occlusion_reasoning	3	0.76563	0.71250	0.20000	0.58270
distractor_contact_shift	minus_occlusion_reasoning	4	0.77186	0.63750	0.13750	0.55339
distractor_contact_shift	minus_occlusion_reasoning	5	0.77246	0.67500	0.16250	0.57245
distractor_contact_shift	minus_occlusion_reasoning	6	0.76998	0.57500	0.20000	0.48657
distractor_contact_shift	minus_occlusion_reasoning	7	0.77348	0.65000	0.07500	0.58642
distractor_contact_shift	minus_occlusion_reasoning	8	0.77627	0.55000	0.13750	0.49466
distractor_contact_shift	minus_occlusion_reasoning	9	0.75996	0.71250	0.10000	0.61821
distractor_contact_shift	minus_passive_flow_factor	0	0.70974	0.58750	0.11250	0.48437
distractor_contact_shift	minus_passive_flow_factor	1	0.71278	0.60000	0.13750	0.48823
distractor_contact_shift	minus_passive_flow_factor	2	0.68676	0.52500	0.20000	0.39516
distractor_contact_shift	minus_passive_flow_factor	3	0.71572	0.57500	0.20000	0.44681
distractor_contact_shift	minus_passive_flow_factor	4	0.70965	0.52500	0.12500	0.43428
distractor_contact_shift	minus_passive_flow_factor	5	0.72028	0.56250	0.16250	0.45462
distractor_contact_shift	minus_passive_flow_factor	6	0.72566	0.58750	0.15000	0.47706
distractor_contact_shift	minus_passive_flow_factor	7	0.71528	0.43750	0.15000	0.36925
distractor_contact_shift	minus_passive_flow_factor	8	0.72547	0.58750	0.15000	0.47700
distractor_contact_shift	minus_passive_flow_factor	9	0.70214	0.60000	0.10000	0.49413
distractor_contact_shift	minus_uncertainty_calibration	0	0.77329	0.62500	0.10000	0.55634
distractor_contact_shift	minus_uncertainty_calibration	1	0.75868	0.68750	0.28750	0.52599
distractor_contact_shift	minus_uncertainty_calibration	2	0.75971	0.55000	0.18750	0.46472
distractor_contact_shift	minus_uncertainty_calibration	3	0.75575	0.61250	0.12500	0.53113
distractor_contact_shift	minus_uncertainty_calibration	4	0.77056	0.58750	0.25000	0.47313
distractor_contact_shift	minus_uncertainty_calibration	5	0.76544	0.65000	0.17500	0.54389
distractor_contact_shift	minus_uncertainty_calibration	6	0.75933	0.70000	0.16250	0.57963
distractor_contact_shift	minus_uncertainty_calibration	7	0.76258	0.58750	0.10000	0.52541
distractor_contact_shift	minus_uncertainty_calibration	8	0.76578	0.63750	0.16250	0.53954

Table 11: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	F1	Success	Unsafe	Utility
distractor_contact_shift	minus_uncertainty_calibration	9	0.75132	0.66250	0.16250	0.55225
distractor_contact_shift	old_v4_causal_score	0	0.78926	0.61250	0.13750	0.54275
distractor_contact_shift	old_v4_causal_score	1	0.78075	0.65000	0.16250	0.55540
distractor_contact_shift	old_v4_causal_score	2	0.78401	0.57500	0.10000	0.52680
distractor_contact_shift	old_v4_causal_score	3	0.77983	0.58750	0.12500	0.52605
distractor_contact_shift	old_v4_causal_score	4	0.78437	0.58750	0.07500	0.54409
distractor_contact_shift	old_v4_causal_score	5	0.78672	0.73750	0.11250	0.63829
distractor_contact_shift	old_v4_causal_score	6	0.78423	0.61250	0.13750	0.53905
distractor_contact_shift	old_v4_causal_score	7	0.78626	0.60000	0.10000	0.54668
distractor_contact_shift	old_v4_causal_score	8	0.78703	0.50000	0.07500	0.48478
distractor_contact_shift	old_v4_causal_score	9	0.77407	0.60000	0.18750	0.51151

F FIXED-RISK SEED METRICS

Table 12: Seed-level fixed-risk metrics.

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
combined_hard_shift	0.02	Calibrated tree	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Causal flow v5	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	7	0.01562	0.01562	1.00000	0.00000
combined_hard_shift	0.02	Oracle	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	1	0.00000	0.00000	0.00000	0.00000

Table 12: Seed-level fixed-risk metrics. (continued)

[illegible]

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
combined_hard_shift	0.10	Calibrated tree	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Causal flow v5	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Oracle	0	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.10	Oracle	1	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.10	Oracle	2	1.00000	0.98438	0.98438	0.01562
combined_hard_shift	0.10	Oracle	3	1.00000	0.98438	0.98438	0.01562
combined_hard_shift	0.10	Oracle	4	1.00000	0.96875	0.96875	0.01562
combined_hard_shift	0.10	Oracle	5	1.00000	1.00000	1.00000	0.00000
combined_hard_shift	0.10	Oracle	6	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.10	Oracle	7	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.10	Oracle	8	1.00000	0.95312	0.95312	0.03125
combined_hard_shift	0.10	Oracle	9	1.00000	0.98438	0.98438	0.01562
combined_hard_shift	0.10	World model	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.20	Calibrated tree	0	0.29688	0.23438	0.78947	0.05263
combined_hard_shift	0.20	Calibrated tree	1	0.26562	0.21875	0.82353	0.05882
combined_hard_shift	0.20	Calibrated tree	2	0.28125	0.17188	0.61111	0.27778
combined_hard_shift	0.20	Calibrated tree	3	0.34375	0.21875	0.63636	0.04545
combined_hard_shift	0.20	Calibrated tree	4	0.25000	0.20312	0.81250	0.12500
combined_hard_shift	0.20	Calibrated tree	5	0.25000	0.17188	0.68750	0.12500
combined_hard_shift	0.20	Calibrated tree	6	0.26562	0.21875	0.82353	0.17647
combined_hard_shift	0.20	Calibrated tree	7	0.31250	0.28125	0.90000	0.00000
combined_hard_shift	0.20	Calibrated tree	8	0.25000	0.18750	0.75000	0.00000
combined_hard_shift	0.20	Calibrated tree	9	0.23438	0.17188	0.73333	0.13333
combined_hard_shift	0.20	Causal flow v5	0	0.32812	0.29688	0.90476	0.09524
combined_hard_shift	0.20	Causal flow v5	1	0.28125	0.21875	0.77778	0.05556
combined_hard_shift	0.20	Causal flow v5	2	0.35938	0.26562	0.73913	0.00000
combined_hard_shift	0.20	Causal flow v5	3	0.35938	0.31250	0.86957	0.04348
combined_hard_shift	0.20	Causal flow v5	4	0.23438	0.18750	0.80000	0.00000
combined_hard_shift	0.20	Causal flow v5	5	0.25000	0.20312	0.81250	0.00000
combined_hard_shift	0.20	Causal flow v5	6	0.37500	0.26562	0.70833	0.12500
combined_hard_shift	0.20	Causal flow v5	7	0.39062	0.25000	0.64000	0.04000
combined_hard_shift	0.20	Causal flow v5	8	0.31250	0.21875	0.70000	0.10000
combined_hard_shift	0.20	Causal flow v5	9	0.31250	0.20312	0.65000	0.20000
combined_hard_shift	0.20	Contrastive action	0	0.57812	0.37500	0.64865	0.10811
combined_hard_shift	0.20	Contrastive action	1	0.53125	0.34375	0.64706	0.11765
combined_hard_shift	0.20	Contrastive action	2	0.67188	0.46875	0.69767	0.06977
combined_hard_shift	0.20	Contrastive action	3	0.65625	0.50000	0.76190	0.11905
combined_hard_shift	0.20	Contrastive action	4	0.68750	0.53125	0.77273	0.13636
combined_hard_shift	0.20	Contrastive action	5	0.46875	0.32812	0.70000	0.06667

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
combined_hard_shift	0.20	Contrastive action	6	0.57812	0.37500	0.64865	0.13514
combined_hard_shift	0.20	Contrastive action	7	0.64062	0.48438	0.75610	0.04878
combined_hard_shift	0.20	Contrastive action	8	0.53125	0.31250	0.58824	0.08824
combined_hard_shift	0.20	Contrastive action	9	0.59375	0.40625	0.68421	0.07895
combined_hard_shift	0.20	Learned correlation	0	0.45312	0.31250	0.68966	0.03448
combined_hard_shift	0.20	Learned correlation	1	0.39062	0.21875	0.56000	0.16000
combined_hard_shift	0.20	Learned correlation	2	0.48438	0.35938	0.74194	0.06452
combined_hard_shift	0.20	Learned correlation	3	0.43750	0.28125	0.64286	0.03571
combined_hard_shift	0.20	Learned correlation	4	0.43750	0.34375	0.78571	0.10714
combined_hard_shift	0.20	Learned correlation	5	0.34375	0.20312	0.59091	0.00000
combined_hard_shift	0.20	Learned correlation	6	0.42188	0.26562	0.62963	0.11111
combined_hard_shift	0.20	Learned correlation	7	0.45312	0.32812	0.72414	0.10345
combined_hard_shift	0.20	Learned correlation	8	0.40625	0.29688	0.73077	0.19231
combined_hard_shift	0.20	Learned correlation	9	0.34375	0.26562	0.77273	0.13636
combined_hard_shift	0.20	Oracle	0	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.20	Oracle	1	1.00000	0.96875	0.96875	0.00000
combined_hard_shift	0.20	Oracle	2	1.00000	0.96875	0.96875	0.01562
combined_hard_shift	0.20	Oracle	3	1.00000	0.98438	0.98438	0.00000
combined_hard_shift	0.20	Oracle	4	1.00000	0.98438	0.98438	0.04688
combined_hard_shift	0.20	Oracle	5	1.00000	0.96875	0.96875	0.00000
combined_hard_shift	0.20	Oracle	6	1.00000	0.96875	0.96875	0.01562
combined_hard_shift	0.20	Oracle	7	1.00000	0.95312	0.95312	0.01562
combined_hard_shift	0.20	Oracle	8	1.00000	1.00000	1.00000	0.00000
combined_hard_shift	0.20	Oracle	9	1.00000	0.96875	0.96875	0.00000
combined_hard_shift	0.20	World model	0	0.53125	0.29688	0.55882	0.05882
combined_hard_shift	0.20	World model	1	0.43750	0.32812	0.75000	0.17857
combined_hard_shift	0.20	World model	2	0.48438	0.34375	0.70968	0.06452
combined_hard_shift	0.20	World model	3	0.51562	0.39062	0.75758	0.06061
combined_hard_shift	0.20	World model	4	0.31250	0.25000	0.80000	0.10000
combined_hard_shift	0.20	World model	5	0.39062	0.32812	0.84000	0.12000
combined_hard_shift	0.20	World model	6	0.46875	0.31250	0.66667	0.06667
combined_hard_shift	0.20	World model	7	0.56250	0.42188	0.75000	0.08333
combined_hard_shift	0.20	World model	8	0.50000	0.31250	0.62500	0.12500
combined_hard_shift	0.20	World model	9	0.53125	0.42188	0.79412	0.05882
distractor_contact_shift	0.02	Calibrated tree	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Causal flow v5	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Oracle	0	0.70312	0.67188	0.95556	0.00000
distractor_contact_shift	0.02	Oracle	1	0.67188	0.65625	0.97674	0.00000
distractor_contact_shift	0.02	Oracle	2	0.70312	0.70312	1.00000	0.00000

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
distractor_contact_shift	0.02	Oracle	3	0.79688	0.78125	0.98039	0.00000
distractor_contact_shift	0.02	Oracle	4	0.81250	0.81250	1.00000	0.00000
distractor_contact_shift	0.02	Oracle	5	0.70312	0.68750	0.97778	0.00000
distractor_contact_shift	0.02	Oracle	6	0.73438	0.71875	0.97872	0.00000
distractor_contact_shift	0.02	Oracle	7	0.64062	0.62500	0.97561	0.07317
distractor_contact_shift	0.02	Oracle	8	0.62500	0.62500	1.00000	0.00000
distractor_contact_shift	0.02	Oracle	9	0.67188	0.65625	0.97674	0.00000
distractor_contact_shift	0.02	World model	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Causal flow v5	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Oracle	0	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.05	Oracle	1	1.00000	0.96875	0.96875	0.03125
distractor_contact_shift	0.05	Oracle	2	1.00000	0.90625	0.90625	0.04688
distractor_contact_shift	0.05	Oracle	3	1.00000	0.95312	0.95312	0.01562
distractor_contact_shift	0.05	Oracle	4	1.00000	1.00000	1.00000	0.03125
distractor_contact_shift	0.05	Oracle	5	1.00000	0.93750	0.93750	0.03125
distractor_contact_shift	0.05	Oracle	6	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.05	Oracle	7	1.00000	0.98438	0.98438	0.03125
distractor_contact_shift	0.05	Oracle	8	1.00000	0.98438	0.98438	0.01562
distractor_contact_shift	0.05	Oracle	9	1.00000	0.96875	0.96875	0.00000
distractor_contact_shift	0.05	World model	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	9	0.00000	0.00000	0.00000	0.00000

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
distractor_contact_shift	0.10	Calibrated tree	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Causal flow v5	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Oracle	0	1.00000	0.98438	0.98438	0.04688
distractor_contact_shift	0.10	Oracle	1	1.00000	0.96875	0.96875	0.00000
distractor_contact_shift	0.10	Oracle	2	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.10	Oracle	3	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.10	Oracle	4	1.00000	0.96875	0.96875	0.00000
distractor_contact_shift	0.10	Oracle	5	1.00000	1.00000	1.00000	0.00000
distractor_contact_shift	0.10	Oracle	6	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.10	Oracle	7	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.10	Oracle	8	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.10	Oracle	9	1.00000	1.00000	1.00000	0.00000
distractor_contact_shift	0.10	World model	0	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	1	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	2	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	3	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	4	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	5	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	6	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	7	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	8	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	9	0.00000	0.00000	0.00000	0.00000
distractor_contact_shift	0.20	Calibrated tree	0	0.23438	0.20312	0.86667	0.06667
distractor_contact_shift	0.20	Calibrated tree	1	0.26562	0.23438	0.88235	0.05882
distractor_contact_shift	0.20	Calibrated tree	2	0.31250	0.25000	0.80000	0.30000
distractor_contact_shift	0.20	Calibrated tree	3	0.26562	0.20312	0.76471	0.00000
distractor_contact_shift	0.20	Calibrated tree	4	0.31250	0.20312	0.65000	0.05000
distractor_contact_shift	0.20	Calibrated tree	5	0.37500	0.26562	0.70833	0.04167
distractor_contact_shift	0.20	Calibrated tree	6	0.29688	0.25000	0.84211	0.05263
distractor_contact_shift	0.20	Calibrated tree	7	0.23438	0.20312	0.86667	0.00000
distractor_contact_shift	0.20	Calibrated tree	8	0.37500	0.29688	0.79167	0.00000
distractor_contact_shift	0.20	Calibrated tree	9	0.21875	0.14062	0.64286	0.14286
distractor_contact_shift	0.20	Causal flow v5	0	0.31250	0.23438	0.75000	0.05000
distractor_contact_shift	0.20	Causal flow v5	1	0.21875	0.17188	0.78571	0.21429
distractor_contact_shift	0.20	Causal flow v5	2	0.26562	0.20312	0.76471	0.05882
distractor_contact_shift	0.20	Causal flow v5	3	0.25000	0.18750	0.75000	0.06250
distractor_contact_shift	0.20	Causal flow v5	4	0.31250	0.26562	0.85000	0.20000
distractor_contact_shift	0.20	Causal flow v5	5	0.40625	0.31250	0.76923	0.07692
distractor_contact_shift	0.20	Causal flow v5	6	0.34375	0.26562	0.77273	0.04545

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Exec. success	False-safe
distractor_contact_shift	0.20	Causal flow v5	7	0.21875	0.17188	0.78571	0.07143
distractor_contact_shift	0.20	Causal flow v5	8	0.34375	0.29688	0.86364	0.00000
distractor_contact_shift	0.20	Causal flow v5	9	0.14062	0.09375	0.66667	0.22222
distractor_contact_shift	0.20	Contrastive action	0	0.53125	0.34375	0.64706	0.17647
distractor_contact_shift	0.20	Contrastive action	1	0.46875	0.35938	0.76667	0.06667
distractor_contact_shift	0.20	Contrastive action	2	0.57812	0.40625	0.70270	0.02703
distractor_contact_shift	0.20	Contrastive action	3	0.53125	0.32812	0.61765	0.05882
distractor_contact_shift	0.20	Contrastive action	4	0.50000	0.39062	0.78125	0.12500
distractor_contact_shift	0.20	Contrastive action	5	0.62500	0.53125	0.85000	0.02500
distractor_contact_shift	0.20	Contrastive action	6	0.48438	0.43750	0.90323	0.12903
distractor_contact_shift	0.20	Contrastive action	7	0.42188	0.29688	0.70370	0.07407
distractor_contact_shift	0.20	Contrastive action	8	0.57812	0.42188	0.72973	0.02703
distractor_contact_shift	0.20	Contrastive action	9	0.50000	0.39062	0.78125	0.12500
distractor_contact_shift	0.20	Learned correlation	0	0.32812	0.29688	0.90476	0.09524
distractor_contact_shift	0.20	Learned correlation	1	0.37500	0.32812	0.87500	0.25000
distractor_contact_shift	0.20	Learned correlation	2	0.40625	0.35938	0.88462	0.19231
distractor_contact_shift	0.20	Learned correlation	3	0.40625	0.37500	0.92308	0.07692
distractor_contact_shift	0.20	Learned correlation	4	0.32812	0.18750	0.57143	0.00000
distractor_contact_shift	0.20	Learned correlation	5	0.45312	0.35938	0.79310	0.17241
distractor_contact_shift	0.20	Learned correlation	6	0.35938	0.26562	0.73913	0.08696
distractor_contact_shift	0.20	Learned correlation	7	0.31250	0.25000	0.80000	0.05000
distractor_contact_shift	0.20	Learned correlation	8	0.50000	0.31250	0.62500	0.06250
distractor_contact_shift	0.20	Learned correlation	9	0.23438	0.15625	0.66667	0.06667
distractor_contact_shift	0.20	Oracle	0	1.00000	0.93750	0.93750	0.00000
distractor_contact_shift	0.20	Oracle	1	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.20	Oracle	2	1.00000	0.95312	0.95312	0.01562
distractor_contact_shift	0.20	Oracle	3	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.20	Oracle	4	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.20	Oracle	5	1.00000	0.98438	0.98438	0.01562
distractor_contact_shift	0.20	Oracle	6	1.00000	0.98438	0.98438	0.01562
distractor_contact_shift	0.20	Oracle	7	1.00000	0.98438	0.98438	0.00000
distractor_contact_shift	0.20	Oracle	8	1.00000	0.96875	0.96875	0.01562
distractor_contact_shift	0.20	Oracle	9	1.00000	0.95312	0.95312	0.01562
distractor_contact_shift	0.20	World model	0	0.43750	0.34375	0.78571	0.10714
distractor_contact_shift	0.20	World model	1	0.45312	0.35938	0.79310	0.03448
distractor_contact_shift	0.20	World model	2	0.42188	0.31250	0.74074	0.00000
distractor_contact_shift	0.20	World model	3	0.39062	0.32812	0.84000	0.16000
distractor_contact_shift	0.20	World model	4	0.46875	0.37500	0.80000	0.00000
distractor_contact_shift	0.20	World model	5	0.56250	0.45312	0.80556	0.13889
distractor_contact_shift	0.20	World model	6	0.37500	0.32812	0.87500	0.12500
distractor_contact_shift	0.20	World model	7	0.37500	0.31250	0.83333	0.08333
distractor_contact_shift	0.20	World model	8	0.46875	0.42188	0.90000	0.06667
distractor_contact_shift	0.20	World model	9	0.31250	0.28125	0.90000	0.15000

G FIXED-RISK PAIRWISE METRICS

Table 13: Fixed-risk paired seed differences for v5.

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
combined_hard_shift	0.02	Learned correlation	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.02	World model	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	coverage	-0.00156	0.00306	-0.00462
combined_hard_shift	0.02	Oracle	fixed_risk_success	-0.00156	0.00306	-0.00462
combined_hard_shift	0.02	Oracle	executed_success	-0.10000	0.19600	-0.29600
combined_hard_shift	0.02	Oracle	false_safe_rate	0.00000	0.00000	0.00000

Table 13: Fixed-risk paired seed differences for v5. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
combined_hard_shift	0.05	Learned correlation	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned correlation	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Calibrated tree	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Calibrated tree	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Contrastive action	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Contrastive action	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.05	World model	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.05	World model	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	World model	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.05	World model	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Oracle	coverage	-1.00000	0.00000	-1.00000
combined_hard_shift	0.05	Oracle	fixed_risk_success	-0.97656	0.00823	-0.98480
combined_hard_shift	0.05	Oracle	executed_success	-0.97656	0.00823	-0.98480
combined_hard_shift	0.05	Oracle	false_safe_rate	-0.00625	0.00500	-0.01125
combined_hard_shift	0.10	Learned correlation	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Calibrated tree	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Calibrated tree	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	coverage	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	fixed_risk_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	executed_success	0.00000	0.00000	0.00000
combined_hard_shift	0.10	World model	false_safe_rate	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Oracle	coverage	-1.00000	0.00000	-1.00000
combined_hard_shift	0.10	Oracle	fixed_risk_success	-0.98125	0.00764	-0.98889
combined_hard_shift	0.10	Oracle	executed_success	-0.98125	0.00764	-0.98889
combined_hard_shift	0.10	Oracle	false_safe_rate	-0.00937	0.00677	-0.01614
combined_hard_shift	0.20	Learned correlation	coverage	-0.09687	0.03021	-0.12709
combined_hard_shift	0.20	Learned correlation	fixed_risk_success	-0.04531	0.03579	-0.08110
combined_hard_shift	0.20	Learned correlation	executed_success	0.07337	0.08515	-0.01178
combined_hard_shift	0.20	Learned correlation	false_safe_rate	-0.02858	0.04076	-0.06934
combined_hard_shift	0.20	Calibrated tree	coverage	0.04531	0.02560	0.01971
combined_hard_shift	0.20	Calibrated tree	fixed_risk_success	0.03437	0.02614	0.00823
combined_hard_shift	0.20	Calibrated tree	executed_success	0.00347	0.09056	-0.08709
combined_hard_shift	0.20	Calibrated tree	false_safe_rate	-0.03352	0.07088	-0.10440
combined_hard_shift	0.20	Contrastive action	coverage	-0.27344	0.04479	-0.31822
combined_hard_shift	0.20	Contrastive action	fixed_risk_success	-0.17031	0.05010	-0.22042
combined_hard_shift	0.20	Contrastive action	executed_success	0.06969	0.06251	0.00718
combined_hard_shift	0.20	Contrastive action	false_safe_rate	-0.03094	0.04286	-0.07380
combined_hard_shift	0.20	World model	coverage	-0.15312	0.02807	-0.18119
combined_hard_shift	0.20	World model	fixed_risk_success	-0.09844	0.03875	-0.13719
combined_hard_shift	0.20	World model	executed_success	0.03502	0.08330	-0.04828
combined_hard_shift	0.20	World model	false_safe_rate	-0.02571	0.05249	-0.07819
combined_hard_shift	0.20	Oracle	coverage	-0.67969	0.03268	-0.71237
combined_hard_shift	0.20	Oracle	fixed_risk_success	-0.73281	0.02718	-0.76000
combined_hard_shift	0.20	Oracle	executed_success	-0.21479	0.05195	-0.26674
combined_hard_shift	0.20	Oracle	false_safe_rate	0.05655	0.04482	0.01173
distractor_contact_shift	0.02	Learned correlation	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	World model	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.02	Oracle	coverage	-0.70625	0.03787	-0.74412

Table 13: Fixed-risk paired seed differences for v5. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
distractor_contact_shift	0.02	Oracle	fixed_risk_success	-0.69375	0.03879	-0.73254
distractor_contact_shift	0.02	Oracle	executed_success	-0.98215	0.00876	-0.99091
distractor_contact_shift	0.02	Oracle	false_safe_rate	-0.00732	0.01434	-0.02166
distractor_contact_shift	0.05	Learned correlation	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	World model	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.05	Oracle	coverage	-1.00000	0.00000	-1.00000
distractor_contact_shift	0.05	Oracle	fixed_risk_success	-0.96719	0.01736	-0.98454
distractor_contact_shift	0.05	Oracle	executed_success	-0.96719	0.01736	-0.98454
distractor_contact_shift	0.05	Oracle	false_safe_rate	-0.02031	0.01026	-0.03057
distractor_contact_shift	0.10	Learned correlation	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Learned correlation	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Calibrated tree	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Contrastive action	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	coverage	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	fixed_risk_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	executed_success	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	World model	false_safe_rate	0.00000	0.00000	0.00000
distractor_contact_shift	0.10	Oracle	coverage	-1.00000	0.00000	-1.00000
distractor_contact_shift	0.10	Oracle	fixed_risk_success	-0.97813	0.00817	-0.98629
distractor_contact_shift	0.10	Oracle	executed_success	-0.97813	0.00817	-0.98629
distractor_contact_shift	0.10	Oracle	false_safe_rate	-0.01094	0.00919	-0.02012
distractor_contact_shift	0.20	Learned correlation	coverage	-0.08906	0.03821	-0.12727
distractor_contact_shift	0.20	Learned correlation	fixed_risk_success	-0.06875	0.05026	-0.11901
distractor_contact_shift	0.20	Learned correlation	executed_success	-0.00244	0.09527	-0.09771
distractor_contact_shift	0.20	Learned correlation	false_safe_rate	-0.00514	0.06549	-0.07063
distractor_contact_shift	0.20	Calibrated tree	coverage	-0.00781	0.02967	-0.03749
distractor_contact_shift	0.20	Calibrated tree	fixed_risk_success	-0.00468	0.02664	-0.03132
distractor_contact_shift	0.20	Calibrated tree	executed_success	-0.00570	0.06020	-0.06590
distractor_contact_shift	0.20	Calibrated tree	false_safe_rate	0.02890	0.06956	-0.04066
distractor_contact_shift	0.20	Contrastive action	coverage	-0.24063	0.03933	-0.27995
distractor_contact_shift	0.20	Contrastive action	fixed_risk_success	-0.17031	0.03608	-0.20639
distractor_contact_shift	0.20	Contrastive action	executed_success	0.02752	0.06226	-0.03475
distractor_contact_shift	0.20	Contrastive action	false_safe_rate	0.01675	0.05124	-0.03448
distractor_contact_shift	0.20	World model	coverage	-0.14531	0.03131	-0.17663
distractor_contact_shift	0.20	World model	fixed_risk_success	-0.13125	0.02337	-0.15462
distractor_contact_shift	0.20	World model	executed_success	-0.05150	0.04880	-0.10030
distractor_contact_shift	0.20	World model	false_safe_rate	0.01361	0.06748	-0.05387
distractor_contact_shift	0.20	Oracle	coverage	-0.71875	0.04810	-0.76685
distractor_contact_shift	0.20	Oracle	fixed_risk_success	-0.75000	0.03954	-0.78954
distractor_contact_shift	0.20	Oracle	executed_success	-0.19447	0.03106	-0.22554
distractor_contact_shift	0.20	Oracle	false_safe_rate	0.09079	0.04975	0.04104

H PRIOR-WORK PRESSURE TABLE

This table is a local prior-work pressure table, not a final curated bibliography. Its role is to keep visible the nearby work that a reviewer could use to challenge novelty or baseline adequacy.

Table 14: Local prior-work pressure sample.

Citation	Title	Year	Venue	Hostile score
(Xu et al., 2019)	Natural teaching for humanoid robot via human-in-the-loop scene-motion cross-modal perception	2019	Industrial Robot: the international journal of robotics research and application arXiv	26.651
(Zhan et al., 2026)	Aerial Manipulation with Contact-Aware Onboard Perception and Hybrid Control	2026		27.250
(Paredes-Valls et al., 2019)	Unsupervised Learning of a Hierarchical Spiking Neural Network for Optical Flow Estimation: From Events to Global Motion Perception	2019	IEEE Transactions on Pattern Analysis and Machine Intelligence	25.810
(Hetherington et al., 2021)	Hey Robot, Which Way Are You Going? Nonverbal Motion Legibility Cues for Human-Robot Spatial Interaction	2021	IEEE Robotics and Automation Letters	24.725
(Scherff & Lappe, 2026)	Flow parsing as causal source separation: A computational model for concurrent retrieval of object and self-motion information from optic flow			25.700
(Aleotti et al., 2020)	Learning End-to-End Scene Flow by Distilling Single Tasks Knowledge	2020		24.607
(Guo et al., 2025)	FlowDreamer: A RGB-D World Model with Flow-based Motion Representations for Robot Manipulation	2025	arXiv	25.550
(Rogers et al., 2017)	Peripheral Visual Cues Contribute to the Perception of Object Movement During Self-Movement	2017	i-Perception	24.517
(Niehorster & Li, 2017)	Accuracy and Tuning of Flow Parsing for Visual Perception of Object Motion During Self-Motion	2017	i-Perception	24.205
(Wang et al., 2024)	DUEL: Depth visUal Ego-motion Learning for autonomous robot obstacle avoidance	2024	The International Journal of Robotics Research	25.823
(Wang et al., 2025)	FlowRAM: Grounding Flow Matching Policy with Region-Aware Mamba Framework for Robotic Manipulation	2025	arXiv	25.550
(Zeng et al., 2026)	FlowHOI: Flow-based Semantics-Grounded Generation of Hand-Object Interactions for Dexterous Robot Manipulation	2026	arXiv	24.550
(Lanillos et al., 2016)	Yielding Self-Perception in Robots Through Sensorimotor Contingencies	2016	IEEE Transactions on Cognitive and Developmental Systems	24.157
(Jauhri et al., 2023)	Active-Perceptive Motion Generation for Mobile Manipulation	2023	arXiv	23.150
(syed Ali Jafar, 2025)	Multimodal Large Language Models (LLMs) for Robotic Manipulation in Unstructured Environments	2025	International Journal of Advanced and Innovative Research (IJAIR)	25.050
(Yang et al., 2026)	Robo3R: Enhancing Robotic Manipulation with Accurate Feed-Forward 3D Reconstruction	2026	arXiv	24.900
(Zaky et al., 2020)	Active Perception and Representation for Robotic Manipulation	2020	arXiv	22.900
(Wang et al., 1998)	Passive compliance versus active compliance in robotbased automated assembly systems	1998	Industrial Robot: An International Journal	22.889
(Duncan et al., 2024)	A Survey of Multimodal Perception Methods for Human-Robot Interaction in Social Environments	2024	ACM Transactions on Human-Robot Interaction	23.448
(Landeck et al., 2024)	Object Motion Manipulation and time perception in virtual reality	2024	Frontiers in Virtual Reality	23.346
(Menze et al., 2015)	JOINT 3D ESTIMATION OF VEHICLES AND SCENE FLOW	2015	ISPRS annals of the photogrammetry, remote sensing and spatial information sciences	22.344
(Wang et al., 2022)	SFGAN: Unsupervised Generative Adversarial Learning of 3D Scene Flow from the 3D Scene Self	2022	Advanced Intelligent Systems	22.195
(Teed & Deng, 2021)	RAFT-3D: Scene Flow using Rigid-Motion Embeddings	2021	2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	22.159
(Cheng et al., 2021)	Human-robot Interaction Method Combining Human Pose Estimation and Motion Intention Recognition	2021		22.028
(Prez-D'Arpino et al., 2024)	Experimental Assessment of Human-Robot Teaming for Multi-step Remote Manipulation with Expert Operators	2024	ACM Transactions on Human-Robot Interaction	22.820
(Wang et al., 2026)	SFGAN: Unsupervised Generative Adversarial Learning of 3D Scene Flow from the 3D Scene Self			21.700
(Cominelli et al., 2021)	Abel: Integrating Humanoid Body, Emotions, and Time Perception to Investigate Social Interaction and Human Cognition	2021	Applied Sciences	21.672
(Fang et al., 2026)	Force Policy: Learning Hybrid Force-Position Control Policy under Interaction Frame for Contact-Rich Manipulation	2026	arXiv	23.550
(Bchner et al., 2026)	Articulated 3D Scene Graphs for Open-World Mobile Manipulation	2026	arXiv	23.550
(Benedicto-Rodríguez et al., 2025)	Understanding Robot Gesture Perception in Children with Autism Spectrum Disorder during Human-Robot Interaction	2025	International Journal of Neural Systems	21.546
(Tremoulet & Feldman, 2000)	Perception of Animacy from the Motion of a Single Object	2000	Perception	21.497
(Mohammad, 2026)	Vision-Based Robotic Manipulation: Grasping in Clutter with Uncertainty-Aware Perception	2026	International Journal of Artificial Intelligence, Data Science and Machine Learning	23.350
(Ghonimy & Abdel-Baky, 2026)	Robotic Fruit Harvesting Systems: Integration of Perception, Manipulation, and Detachment for Autonomous Harvesting	2026	Agronomy	24.350
(Jrges & Harris, 2022)	Object speed perception during lateral visual self-motion	2022	Attention, Perception, & Psychophysics	21.332

Table 14: Local prior-work pressure sample. (continued)

Citation	Title	Year	Venue	Hostile score
(Franceschi et al., 2020)	Precise Robotic Manipulation of Bulky Components	2020	IEEE Access	21.246
(Jiang et al., 2025)	Kaiwu: A Multimodal Manipulation Dataset and Framework for Robot Learning and Human-Robot Interaction	2025	arXiv	22.200

REFERENCES

- Filippo Aleotti, Matteo Poggi, Fabio Tosi, and Stefano Mattoccia. Learning end-to-end scene flow by distilling single tasks knowledge, 2020. openalex; 10.1609/aaai.v34i07.6613.
- Martin Bchner, Adrian Rfer, Tim Engelbracht, Tim Welschhold, Zuria Bauer, Hermann Blum, Marc Pollefeys, and Abhinav Valada. Articulated 3d scene graphs for open-world mobile manipulation, 2026. arXiv; <http://arxiv.org/abs/2602.16356v1>.
- Gema Benedicto-Rodriguez, Facundo Bosch, Carlos G. Juan, Maria Paula Bonomini, Antonio Fernndez-Caballero, Eduardo Fernandez-Jover, and Jose Manuel Ferrndez-Vicente. Understanding robot gesture perception in children with autism spectrum disorder during human-robot interaction, 2025. International Journal of Neural Systems; 10.1142/s0129065725500261.
- Yalin Cheng, Pengfei Yi, Liu Rui, Jing Dong, Dongsheng Zhou, and Qiang Zhang. Human-robot interaction method combining human pose estimation and motion intention recognition, 2021. openalex; 10.1109/cscwd49262.2021.9437772.
- Lorenzo Cominelli, Gustav Hoegen, and Danilo De Rossi. Abel: Integrating humanoid body, emotions, and time perception to investigate social interaction and human cognition, 2021. Applied Sciences; 10.3390/app11031070.
- John A. Duncan, Farshid Alambeigi, and Mitchell W. Pryor. A survey of multimodal perception methods for human-robot interaction in social environments, 2024. ACM Transactions on Human-Robot Interaction; 10.1145/3657030.
- Hongjie Fang, Shirun Tang, Mingyu Mei, Haoxiang Qin, Zihao He, Jingjing Chen, Ying Feng, Chenxi Wang, Wanxi Liu, Zaixing He, Cewu Lu, and Shiquan Wang. Force policy: Learning hybrid force-position control policy under interaction frame for contact-rich manipulation, 2026. arXiv; <http://arxiv.org/abs/2602.22088v2>.
- Paolo Franceschi, Nicola Castaman, Stefano Ghidoni, and Nicola Pedrocchi. Precise robotic manipulation of bulky components, 2020. IEEE Access; 10.1109/access.2020.3043069.
- Mohamed Ghonimy and Nagdy F. Abdel-Baky. Robotic fruit harvesting systems: Integration of perception, manipulation, and detachment for autonomous harvesting, 2026. Agronomy; 10.3390/agronomy16121127.
- Jun Guo, Xiaojian Ma, Yikai Wang, Min Yang, Huaping Liu, and Qing Li. Flowdreamer: A rgb-d world model with flow-based motion representations for robot manipulation, 2025. arXiv; <http://arxiv.org/abs/2505.10075v1>.
- Nicholas J. Hetherington, Elizabeth A. Croft, and H.F. Machiel Van der Loos. Hey robot, which way are you going? nonverbal motion legibility cues for human-robot spatial interaction, 2021. IEEE Robotics and Automation Letters; 10.1109/lra.2021.3068708.
- Snehal Jauhri, Sophie Lueth, and Georgia Chalkatzaki. Active-perceptive motion generation for mobile manipulation, 2023. arXiv; <http://arxiv.org/abs/2310.00433v2>.
- Shuo Jiang, Haonan Li, Ruochen Ren, Yanmin Zhou, Zhipeng Wang, and Bin He. Kaiwu: A multimodal manipulation dataset and framework for robot learning and human-robot interaction, 2025. arXiv; 10.1109/lra.2025.3609615.

- 1998 Björn Jrges and Laurence R. Harris. Object speed perception during lateral visual self-motion, 2022.
 1999 Attention, Perception, & Psychophysics; 10.3758/s13414-021-02372-4.
- 2000
- 2001 Maximilian Landeck, Fabian Unruh, Jean-Luc Lugin, and Marc Erich Latoschik. Object mo-
 2002 tion manipulation and time perception in virtual reality, 2024. *Frontiers in Virtual Reality*;
 2003 10.3389/frvir.2024.1390703.
- 2004 Pablo Lanillos, Emmanuel DeanLeon, and Gordon Cheng. Yielding self-perception in robots
 2005 through sensorimotor contingencies, 2016. *IEEE Transactions on Cognitive and Developmen-
 2006 tal Systems*; 10.1109/tcds.2016.2627820.
- 2007
- 2008 Marco Menze, Christian Heipke, and Andreas Geiger. Joint 3d estimation of vehicles and scene flow,
 2009 2015. *ISPRS annals of the photogrammetry, remote sensing and spatial information sciences*;
 2010 10.5194/isprsannals-ii-3-w5-427-2015.
- 2011 Rasool Javeed Mohammad. Vision-based robotic manipulation: Grasping in clutter with
 2012 uncertainty-aware perception, 2026. *International Journal of Artificial Intelligence, Data Science
 2013 and Machine Learning*; 10.63282/3050-9262.ijaidsmi-v7i1p150.
- 2014
- 2015 Diederick C. Niehorster and Li Li. Accuracy and tuning of flow parsing for visual perception of
 2016 object motion during self-motion, 2017. *i-Perception*; 10.1177/2041669517708206.
- 2017
- 2018 Federico Paredes-Valls, Kirk Y. W. Scheper, and Guido de Croon. Unsupervised learning of a
 2019 hierarchical spiking neural network for optical flow estimation: From events to global mo-
 2020 tion perception, 2019. *IEEE Transactions on Pattern Analysis and Machine Intelligence*;
 2021 10.1109/tpami.2019.2903179.
- 2022 Claudia Prez-D’Arpino, Rebecca P. Khurshid, and Julie A. Shah. Experimental assessment of
 2023 human-robot teaming for multi-step remote manipulation with expert operators, 2024. *ACM
 2024 Transactions on Human-Robot Interaction*; 10.1145/3618258.
- 2025
- 2026 Cassandra Rogers, Simon K. Rushton, and Paul A. Warren. Peripheral visual cues con-
 2027 tribute to the perception of object movement during self-movement, 2017. *i-Perception*;
 2028 10.1177/2041669517736072.
- 2029
- 2030 Malte Scherff and Markus Lappe. Flow parsing as causal source separation: A computational model
 2031 for concurrent retrieval of object and self-motion information from optic flow, 2026. *crossref*;
 2032 10.1101/2024.11.04.621864.
- 2033 Naqvi syed Ali Jafar. Multimodal large language models (llms) for robotic manipulation in unstruc-
 2034 tured environments, 2025. *International Journal of Advanced and Innovative Research (IJAIR)*;
 2035 10.63056/ijair.1.3.2025.100.
- 2036 Zachary Teed and Jia Deng. Raft-3d: Scene flow using rigid-motion embeddings,
 2037 2021. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*;
 2038 10.1109/cvpr46437.2021.00827.
- 2039
- 2040 Patrice D Tremoulet and Jacob Feldman. Perception of animacy from the motion of a single object,
 2041 2000. *Perception*; 10.1068/p3101.
- 2042
- 2043 Guangming Wang, Chaokang Jiang, Zehang Shen, Yanzi Miao, and Hesheng Wang. Sfgan: Unsu-
 2044 pervised generative adversarial learning of 3d scene flow from the 3d scene self, 2022. *Advanced
 2045 Intelligent Systems*; 10.1002/aisy.202100197.
- 2046
- 2047 Guangming Wang, Chaokang Jiang, Zehang Shen, Yanzi Miao, and Hesheng Wang. Sfgan: Unsu-
 2048 pervised generative adversarial learning of 3d scene flow from the 3d scene self, 2026. *crossref*;
 2049 10.22541/au.163335790.03073492/v1.
- 2050
- 2051 Naiyao Wang, Bo Zhang, Haixu Chi, Hua Wang, Sen McLoone, and Hongbo Liu. Duel: Depth
 visual ego-motion learning for autonomous robot obstacle avoidance, 2024. *The International
 Journal of Robotics Research*; 10.1177/02783649231210325.

- Sen Wang, Le Wang, Sanping Zhou, Jingyi Tian, Jiayi Li, Haowen Sun, and Wei Tang. Flowram: Grounding flow matching policy with region-aware mamba framework for robotic manipulation, 2025. arXiv; <http://arxiv.org/abs/2506.16201v1>.
- Wei Wang, Robert N.K. Loh, and Edward Y. Gu. Passive compliance versus active compliance in robotbased automated assembly systems, 1998. Industrial Robot: An International Journal; 10.1108/01439919810196964.
- Wenbin Xu, Xudong Li, Liang Gong, Yixiang Huang, Zeyuan Zheng, Zelin Zhao, Lujie Zhao, Binhao Chen, Haozhe Yang, Li Cao, and Chengliang Liu. Natural teaching for humanoid robot via human-in-the-loop scene-motion cross-modal perception, 2019. Industrial Robot: the international journal of robotics research and application; 10.1108/ir-06-2018-0118.
- Sizhe Yang, Linning Xu, Hao Li, Juncheng Mu, Jia Zeng, Dahua Lin, and Jiangmiao Pang. Robo3r: Enhancing robotic manipulation with accurate feed-forward 3d reconstruction, 2026. arXiv; <http://arxiv.org/abs/2602.10101v2>.
- Youssef Zaky, Gaurav Paruthi, Bryan Tripp, and James Bergstra. Active perception and representation for robotic manipulation, 2020. arXiv; <http://arxiv.org/abs/2003.06734v1>.
- Huajian Zeng, Lingyun Chen, Jiaqi Yang, Yuntai Zhang, Fan Shi, Peidong Liu, and Xingxing Zuo. Flowhoi: Flow-based semantics-grounded generation of hand-object interactions for dexterous robot manipulation, 2026. arXiv; <http://arxiv.org/abs/2602.13444v1>.
- Yuanzhu Zhan, Yufei Jiang, Muqing Cao, and Junyi Geng. Aerial manipulation with contact-aware onboard perception and hybrid control, 2026. arXiv; <http://arxiv.org/abs/2602.08251v2>.